MALAWI PRIMARY EDUCATION

Mathematics

Teachers' guide for Standard 7

Malawi Institute of Education

Contents

About this book

Rationale for mathematics

Mathematics aims at developing the learners' critical awareness of how mathematical relationships are used in social, environmental, cultural and economic context.

At an early stage, the learners will be able to count and carry out basic mathematical operations. At a later stage, the learners will be able to make inferences using manipulated data and apply their mathematical knowledge to solve practical
problems in daily life.

Core elements and their outcomes

This teachers' guide has six core elements. The following are the core elements and their intended outcomes:

Numbers, operations and relationships

The learner will be able to use numbers and their relationships to solve practical problems.

Accounting and business studies

The learner will be able to use simple accounting procedures that will enhance decision making in business and private enterprise.

Space and shape

The learner will be able to describe characteristics of space and shape and their application in everyday life.

Measurement

The learner will be able to use appropriate measurement concepts and skills in real-life situations.

Patterns, functions and algebra

The learner will be able to use algebraic language and skills to solve textual problems.

Data handling

The learner will be able to analyse and interpret data for decision making by using graphs, tables and models.

Unique features of the teachers' guide

This mathematics teachers' guide is based on both its related learner's book and the mathematics syllabus. The developers of this book had the learner and the teacher in mind when they were developing the activities. The development of each topic is based on the principle of moving from the known to the unknown, and simple to complex, and even concrete to semi-concrete.

UNIT 1 Counting

Time allocation 12 lessons

Introduction

Learners have heard about or seen large numbers being used in situations such as budget estimates and world populations. However, it is difficult for some of them to either name or write such numbers.

In this unit, learners will be introduced to reading and writing numbers in figures and words up to 1,000,000,000 (one billion).

Success criteria

Learners must be able to:

- read numbers in figures and words up to one billion
- write numbers in figures and words up to one billion

Developmental areas

Skills

Ensure that the learners acquire the following skills:

- counting in intervals
- arranging numbers in ascending and descending order

Concepts and knowledge

Ensure that the learners understand and acquire the following concepts and knowledge:

- names of numbers
- numbers in figures and words up to one billion
- ascending and descending order of numbers up to one billion

Attitudes and values

Ensure that the learners appreciate the importance of large numbers.

Special needs education

Ensure that the activities are adapted for learners with special educational needs. This will help them develop their potentials and become independent citizens. Observe and record their performance.

Background information

Numbers up to 10,000,000 have been introduced to learners in the previous classes. It is hoped that learners will find it easier to learn numbers bigger than 10,000,000. It is important to find out how much the learners know before proceeding with this unit.

In this unit, learners will be introduced to numbers up to one billion.

Activity 1 Reading numbers from 10,000,000 to 100,000,000

Time allocation 2 lessons

Suggested teaching, learning and assessment resources

Number cards, checklist, braille, raised number cards, mirror, feathers.

Instructions

1 Ask the learners to read numbers in 10 millions from 10,000,000 to 100,000,000 as shown below.

10,000,000	20,000,000	30,000,000	40,000,000	50,000,000
60,000,000	70,000,000	80,000 000	90,000,000	100,000,000

2 Let the learners copy the numbers in their exercise books.

3 Let them read numbers in intervals of one million eg numbers in the range of 90,000,000 to 95,000,000 are read as:

90,000,000	ninety million
91,000,000	ninety one million
92,000,000	ninety two million
93,000,000	ninety three million
94,000,000	ninety four million
95,000,000	ninety five million

4 Help learners to write down the numerals in their exercise books.

5 Repeat instruction 3 using other intervals such as 2,500,000 and 5,000,000.

6 Discuss the example in the learners' book.

7 Let the learners do exercise 1A from the learners' book.

Activity 2 Counting in different intervals up to 1,000,000,000

Time allocation 2 lessons

Suggested teaching, learning and assessment resources

Chart showing numbers from 100,000,000 to 1,000,000,000, checklist, braille, raised number cards, mirror, feathers.

Instructions

1 Display a chart showing numbers from 100,000,000 to 1,000,000,000 in intervals of 100,000,000 as shown below:

100,000,000	one hundred million
200,000,000	two hundred million
300,000,000	three hundred million
400,000,000	four hundred million
500,000,000	five hundred million
600,000,000	six hundred million
700,000,000	seven hundred million
800,000,000	eight hundred million
900,000,000	nine hundred million
1,000,000,000	one billion

2 Let the learners read the numbers in ascending and descending order.

3 Help learners to read and write numbers from 100,000,000 to 1,000,000,000 in intervals of 100,000,000.

4 Repeat steps 1 to 3 using other intervals such as 25,000,000 and 50,000,000

5 Let the learners do Exercise 1B from the learners' book.

Activity 3 Arranging numbers in ascending order

Time allocation 2 lessons

Suggested teaching, learning and assessment resources

Number cards, checklist, braille, raised number cards, mirror, feathers.

Instructions

1 Let the learners be in groups.

2 Distribute cards bearing the following numbers: 100,000,000; 120,000,000; 140,000,000; 160,000,000; 180,000,000; 200,000,000; 220,000,000.

3 Let each group arrange and display the numbers in ascending order.

4 Discuss with learners to establish the correct order of arranging the numbers.

5 Repeat instructions 2 to 4 using different sets of numbers.
6 Discuss the example in the learners' book.
7 Let learners do exercise 1C from the learners' book.

Activity 4 Arranging numbers in descending order

Time allocation 2 lessons

Suggested teaching, learning and assessment resources

Number cards, checklist, braille, raised number cards, mirror, feathers.

Instructions

1 Let the learners to be in groups.
2 Distribute cards bearing the following numbers: 512,000,000; 556,000,000; 610,000,000; 650,000,000; 740,000,000; 750,000,000; 880,000,000.
3 Let learners in each group arrange and display the numbers in descending order.
4 Discuss with learners to establish the correct order of arranging the numbers.
5 Repeat instructions 2 to 4 using different sets of numbers.
6 Discuss the example in the learners' book.
7 Let learners do exercise 1D from the learners' book.

Activity 5 Writing numbers in words

Time allocation 2 lessons

Suggested teaching, learning and assessment resources

A chart showing numbers, checklist, braille, raised number cards, mirror, feathers.

Instructions

1 Let the learners be in groups.
2 Ask the learners to name the numbers on the chart eg 180,000,000; 505,750,000; 816,078,102.
3 Help learners to write the numbers in words, ie 816,078,102 is written as eight hundred and sixteen million, seventy eight thousand one hundred and two.
4 Repeat instructions 2 and 3 using other numbers.
5 Discuss the example in the learners' book.
6 Let learners do Exercise 1E from the learners' book.

Activity 6 Writing numbers in figures

Time allocation 2 lessons

Suggested teaching, learning and assessment resources

Chart showing numbers in words, checklist, braille, raised number cards, mirror, feathers.

Instructions

1 Let the learners be in groups
2 Ask the learners to read the numbers on the chart.
3 Help learners write the number in figures as follows:

Four hundred and eighteen million	418,000,000
Seven hundred and five thousand	705,000
Three hundred and eighteen	318
	418,705,318

4 Discuss the example in the learners' book.
5 Let learners do exercise 1F from the learners' book.

Assessment

Use the table below to assess each learner.

Name of the learner: ______________________________

Is the learner able to:	Yes	No
1 read numbers from 10,000,000 to one billion?		
2 count in different intervals up to one billion?		
3 arrange numbers in ascending order?		
4 arrange numbers in descending order?		
5 write numbers in words?		
6 write numbers in figures?		

UNIT 2 Roman numerals

Time allocation 4 lessons

Introduction

By now learners are able to read and write Hindu-Arabic numerals. These are numbers such as 1, 2, 3, 4, 5. In real life situations, Roman numerals are also used. For example they are used on some clock faces, wrist watches and preliminary pages of some books. In this unit learners will be introduced to Roman numerals up to twenty.

Success criteria

Learners must be able to recognise Roman numerals.

Developmental areas

Skills

Ensure that the learners acquire the following skills:

- identifying Roman numerals
- reading Roman numerals
- writing Roman numerals
- ordering Roman numerals

Concepts and knowledge

Ensure that the learners understand and acquire the concept of Roman numerals.

Attitudes and values

Ensure that learners appreciate the importance of Roman numerals.

Special needs education

Ensure that the activities are adapted for learners with special educational needs. This will help them develop their potentials and become independent citizens. Observe and record their performance.

Background information

In this class learners will make use of three basic Roman numerals. These are I, V and X. The symbol I stands for one, V stands for five and X stands for ten. It is important to note that in Roman numerals there is no symbol for zero.

When building up numbers using the Roman numerals the following rules are followed:

Rule 1

When the symbol is repeated the value is also repeated except for V.

Examples

II = 2

XX= 10 + 10 = 20

Rule 2

A numeral should not be repeated more than three times eg III is correct while IIII is not.

Rule 3

When I is to the right of either V or X it shows that the number is one more than V or X

Examples

XI	=	11
VI	=	6
or		
XIII	=	13
VII	=	7

Rule 4

When I is to the left of either V or X it shows that the number is one less than V or X

Examples

IX	=	9
IV	=	4

Note: In some cases two rules are applied to write Roman numerals.

For example XIX = 19

Activity 1 Introducing Roman numerals up to 20

Time allocation 2 lessons

Suggested teaching, learning and assessment resources

Objects with Roman numerals eg wall clock, wrist watches, raised clock faces, clock faces, number cards, raised number cards, braille, mirror, feathers.

Instructions

1. Let learners be in groups.
2. Distribute to each group clock faces with Roman numerals and others with Hindu–Arabic numerals.
3. Ask learners to identify the differences on the numerals used.
4. Discuss with learners the two types of numerals.
5. Assist learners to establish that Roman numerals use the letters of the alphabet.
6. Let learners do exercise 2A from the learners' book.

Activity 2 Writing Roman numerals up to 20

Time allocation 2 lessons

Suggested teaching, learning and assessment resources

Objects with Roman numerals eg wall clock faces, clock faces, number cards, raised number cards, wrist watches, checklist, braille, raisedclock faces, mirror, feathers.

Instructions

1. Let learners be in groups.
2. Distribute to each group, clock faces with Hindu-Arabic and other with Roman numerals.
3. Ask learners to relate Hindu-Arabic numeral to Roman numerals using the clock faces eg:

Hindu-Arabic numerals	Roman numerals
1	I
2	II
3	III
4	IV
5	V
6	VI
7	VII
8	VIII
9	IX
10	X
11	XI
12	XII

4. Ask learners to identify the letters of the alphabet used.
5. Discuss with learners how other Roman numerals are developed eg 13 is XIII; 14 is XIV; 17 is XVII

6 Help learners to establish that when writing roman numerals after 10, the bigger number is written first eg 13 is written as XIII and not as IIIX.

7 Discuss the examples in exercise 2B and 2C in the learners' book.

8 Let learners do exercises 2B and 2C from the learners' book.

Activity 3 Arranging Roman numerals in ascending and descending order

Time allocation 1 lesson

Suggested teaching, learning and assessment resources

Number cards, raised number cards, checklist, braille, mirror, feathers.

Instructions

1 Let learners be in groups.

2 Distribute number cards with the following numbers to each group: (VII, VIII, IX, X, XI, XII.)

3 Let learners arrange the numbers in ascending order.

4 Ask learners to present their work to the class.

5 Repeat instructions 2 to 4 for descending order.

6 Discuss the examples in exercises 2D and 2E in the learners' book.

7 Let learners do exercise 2D and 2E from the learners' book.

Assessment

Use the table below to assess each learner.

Name of the learner: ___________________________

Is the learner able to:	Yes	No
1 identify roman numerals up to 20?		
2 read Roman numerals up to 20?		
3 write Roman numerals up to 20?		
4 order Roman numerals up to 20?		

UNIT 2 Basic operations on whole numbers

Time allocation 16 lessons

Introduction

In the previous classes, learners carried out the basic operations on whole numbers separately. However, in real life situations, there are cases where learners may need to use two or more basic operations at the same time. Therefore it is necessary to equip learners with the skills of using several operations in the same problem. In this unit, learners will be carrying out two or more operations in the same problem. They will also be solving practical problems involving two or more basic operations.

Success criteria

Learners must be able to:

- add and subtract whole numbers in the same problem
- multiply and divide whole numbers in the same problem
- solve practical problems on whole numbers involving basic operations

Developmental areas

Skills

Ensure that the learners acquire the following skills:

- adding and subtracting whole numbers in the same problem up to three terms
- multiplying and dividing whole numbers in the same problem up to three terms
- adding and multiplying whole numbers in the same problem
- adding and dividing whole numbers in the same problem
- subtracting and multiplying numbers in the same problem
- subtracting and dividing whole numbers in the same problem
- solving practical problems that involve two or more basic operations on whole numbers in the same problem
- solving problems involving adding, subtracting, multiplying and dividing of numbers in the same problem

Concepts and knowledge

Ensure that the learners understand and acquire the concept and knowledge of order of operations.

Attitudes and values

Ensure that the learners appreciate the importance of order of operations when solving problems.

Special needs education

Ensure that the activities are adapted for learners with special educational needs. This will help them develop their potentials and become independent citizens. Observe and record their performance.

Background information

Problems that involve two or more operations are more challenging than those that involve one operation only. However, solving such problems becomes easier if the operations are dealt with one at a time. The following order of carrying out the operations is recommended: division, multiplication, addition and finally subtraction. If a problem has brackets, we deal with the operations in the bracket first before carrying out the other operations.

Activity 1 Adding and subtracting numbers

Time allocation 2 lessons

Suggested teaching, learning and assessment resources

Raised number cards, number cards, braille, mirror, feathers.

Instructions

1 Let learners be in groups.
2 Let each group solve the following problems 79 – 56 + 10 and 105 – 115 + 35.
3 Let learners share their solutions.
4 Discuss with learners the working sequences as follows:

79 - 56 + 10	105 – 115 + 35
= 79 + 10 - 56	= 105 + 35 – 115
= 89 - 56	= 140 – 115
= 33	= 25

5 Assist them to establish that addition is done first then subtraction.
6 Discuss the example in the learners' book.
7 Let the learners do exercise 3A from the learners' book.

Activity 2 Multiplying and dividing numbers

Time allocation 2 lessons

Suggested teaching, learning and assessment resources

Number cards, raised number cards, learners' experiences, braille, mirror, feathers.

Instructions

1 Let learners be in groups.
2 Provide each group with a problem for example: 12 x 40 ÷ 5

3 Ask the learners to solve the problem
4 Let each group report their working sequence.
5 Discuss with learners to establish that starting with either multiplication or division gives the same answer.

$$12 \times 40 \div 5$$
$$= \frac{12 \times 40}{5}$$
$$= \frac{480}{5}$$
$$= 96$$

$$12 \times (40 \div 5)$$
$$= 12 \times 8$$
$$= 96$$

6 Discuss the example in the learners' book.
7 Let the learners do exercise 3B from the learners' book.

Activity 3 Adding and multiplying numbers

Time allocation 2 lessons

Suggested teaching, learning and assessment resources

Number cards, learners' experiences, raised number cards, braille, mirror, feathers.

Instructions

1 Let learners be in groups.
2 Let learners work out the following: $21 + 53 \times 55$
3 Ask them to report how they have come up with the answer.
4 Discuss with learners the working process as follows:

$$21 + 53 \times 55$$
$$= 21 + (53 \times 55)$$
$$= 21 + 2{,}915$$
$$= 2{,}936$$

5 Assist learners to establish that multiplication is done first then addition.
6 Let the learners do exercise 3C from the learners' book.

Activity 4 Adding and dividing numbers

Time allocation 2 lessons

Suggested teaching, learning and assessment resources

Number cards and raised number cards, learners' experiences, braille, mirror, feathers.

Instructions

1 Let learners be in groups.

2 Let learners work out the following:

705 ÷ 47 + 900

3 Ask the learners to describe how they have come up with the answer.

4 Discuss with learners the working process as follows:

$$
\begin{aligned}
& 705 \div 47 + 900 \\
&= (705 \div 47) + 900 \\
&= (705 \div 47) + 900 \\
&= 15 + 900 \\
&= 915
\end{aligned}
$$

5 Assist them to establish that division is done first then addition.

6 Discuss the example in the learners' book.

7 Let learners do exercise 3D from the learners' book.

Activity 5 Subtracting and multiplying numbers

Time allocation 2 lessons

Suggested teaching, learning and assessment resources

Raised number cards, number cards, learners' experiences, braille, mirror, feathers.

Instructions

1 Let learners be in groups.
2 Let learners work out the following: 13 x 3 − 10
3 Ask learners to report how they have come up with the answer.
4 Discuss with learners the working sequence as follows:

$$
\begin{aligned}
& 13 \times 3 - 10 \\
&= (13 \times 3) - 10 \\
&= 39 - 10 \\
&= 29
\end{aligned}
$$

5 Assist learners to establish that multiplication is done first then subtraction.

6 Let the learners do exercise 3E from the learners' book.

Activity 6 Subtracting and dividing numbers

Time allocation 1 lesson

Suggested teaching, learning and assessment resources

Raised number cards, learners' experiences, number cards, braille, mirror, feathers.

Instructions

1 Let learners be in groups.
2 Ask learners to simplify the following: 126 ÷ 3 − 21
3 Let learners present their work.

4 Discuss the working sequence with learners as follows:

126 ÷ 3 − 21

= 42 − 21

= 21

5 Assist them to establish that division is done first then subtraction.

6 Discuss the example in the learners's book.

7 Let the learners do exercise 3F from the learners' book.

Activity 7 Adding, subtracting, multiplying and dividing numbers

Time allocation 2 lessons

Suggested teaching, learning and assessment resources

Number cards, raised number cards, braille, mirror, feathers.

Instructions

1 Let learners be in groups.

2 Let learners work out the following:

200 − 800 ÷ 20 + 6 x 3

3 Ask learners to report how they have come up with the answer.

4 Discuss with learners the working sequence as follows:

200 − 800 ÷ 20 + 6 x 3

= 200 − 40 + 18

= 160 + 18

= 178

5 Assist learners to establish that the operations are done in the following order: brackets, division; multiplication; addition and subtraction.

6 Let the learners do exercise 3G from the learners' book.

Activity 8 Solving practical problems involving basic operations of whole numbers

Time allocation 2 lessons

Suggested teaching, learning and assessment resources

Learners' experiences, problem card, braille, mirror, feathers.

Instructions

1 Let learners be in groups.

2 Ask learners to solve the following problem:

The sum of two numbers is 56. One of the numbers is the quotient of 48 and 4. Find the other number.

3 Let them report how they have solved the problem.

4 Discuss with learners the working sequence as follows:

$56 - 48 \div 4$

$= 56 - (48 \div 4)$

$= 56 - 12$

$= 44$

5 Discuss the example in the learners' book.

6 Let the learners do exercise 3H from learners' book.

Review exercise

Time allocation 1 lesson

Let learners do the review exercise at the end of unit 3.

Assessment

Use the table below to assess each learner.

Name of the learner: ______________________________

How best is the learner able to:	Excellent	Good	Average	Needs support
1 add and subtract numbers?				
2 multiply and divide numbers?				
3 add and multiply numbers?				
4 add and divide numbers?				
5 subtract and multiply numbers?				
6 subtract and divide numbers?				

Averages

Time allocation 5 lessons

Introduction

Average is a concept which is used in everyday life. People talk about average speed, average temperature, average rainfall and even average income. It is therefore important that learners acquire the concept of average because they will use it in many life situations. In this unit, learners will be introduced to averages.

Success criteria

Learners must be able to find averages of numbers.

Developmental areas

Skills

Ensure that learners acquire the skill of finding averages.

Concepts and knowledge

Ensure that learners understand and acquire the concept of average.

Attitudes and values

Ensure that learners appreciate the importance of average.

Special needs education

Ensure that the activities are adapted for learners with special educational needs. This will help them develop their potentials and become independent citizens. Observe and record their performance.

Background information

It is important to introduce the concept of average through the use of concrete examples. For example the average rainfall at a school for one week can be calculated by adding up the rainfall figures for each day and divide the total by 7. In this unit learners will work out average of whole numbers only.

Activity 1 Introducing averages

Time allocation 1 lesson

Suggested teaching, learning and assessment resources

counters, data, charts, checklist, braille, mirror, feathers.

Instructions

1 Ask 4 learners to come in front of the class and give them sticks as follows: John 5, Mary 8, Grace 7 and Peter 4 sticks.
2 Discuss with learners how these sticks can be shared equally among the 4 learners.
3 Let the learners put the sticks together.
4 Let the learners share the sticks equally by picking one stick at a time until all sticks are picked.
5 Ask each learner to count the sticks he/she has picked.
6 Discuss with learners to establish that **average** is the number got by each member after sharing the grouped items; for example 6 is the average.
7 Discuss with learners that the average number of sticks can be found by adding the sticks and dividing by number of learners as follows:

$$\frac{5 + 8 + 7 + 4 \text{ sticks}}{4}$$

$$= \frac{24 \text{ sticks}}{4}$$

$$= 6 \text{ sticks}$$

Activity 2 Finding averages

Time allocation 2 lessons

Suggested teaching, learning and assessment resources

Stones, sticks, learners, checklist, braille, mirror, feathers.

Instructions

1 Let learners be in groups.
2 Ask learners to work out the average of the following numbers: 20, 18, 16 and 14.
3 Ask each group to present their work.
4 Discuss with learners that the average could be worked out as follows:

$$\frac{20 + 18 + 16 + 14}{4}$$

$$= \frac{68}{4}$$

$$= 17$$

5 Let learners do exercise 4A in the learners' book.

Activity 3 Solving practical problems on averages

Time allocation 1 lesson

Suggested teaching, learning and assessment resources

Stones, sticks, books, learners, braille, mirror, feathers.

Instructions

1 Let learners be in groups.
2 Ask learners to work out the following:
The ages of 6 learners are 12 years, 14 years, 10 years, 12 years, 11 years, and 1
years. Find the average age.
3 Let the groups report their findings.
4 Discuss with them to establish that the average age of the 6 learners can be worked out as follows:

$$\frac{12 + 14 + 10 + 12 + 11 + 13}{6}$$

$$= \frac{72}{6}$$

$$= 12 \text{ years}$$

5 Discuss with learners the following example:
"The average length of 5 planks is 105cm and the average of four of them is 85 cm. find the length of the fifth plank."

Total length of 5 planks	=	105cm x 5
	=	525cm
Total length of 4 planks	=	85cm x 4
	=	340cm
Length of fifth plank	=	525cm
	=	− 340cm
		185cm

6 Discuss the example in the learners book.
7 Let learners do exercise 4B from the learners' book.

Assessment

Use the table below to assess each learner.

Name of the learner: ____________________

How best is the learner able to:	Excellent	Good	Average	Needs support
Calculate average?				

UNIT 5 Highest Common Factors (HCF) and Lowest Common Multiples (LCM)

Time allocation 10 lessons

Introduction

In Standard 6, learners were introduced to highest common factors (HCF) using factor, division and continued division methods. They were also introduced to various ways of finding multiples and lowest common multiples (LCM) of numbers. The knowledge of LCM and HCF is important in simplifying common fractions.

In this unit learners will be introduced to lowest common multiples and highest common factors of numbers up to four terms using factor method.

Success criteria

Learners must be able to:

- find Highest Common Factors (HCF) of numbers
- find Lowest Common Multiple (LCM) of numbers

Developmental areas

Skills

Ensure that the learners acquire the following skills:

- finding HCF of numbers using factor method
- finding HCF of numbers using division method
- finding LCM of numbers up to four terms
- finding LCM of numbers using factor method

Concepts and knowledge

Ensure that learners acquire the following concepts and knowledge:

- using factors to find the HCF of numbers
- using division to find the HCF of numbers
- using numbers up to four terms to find LCM of numbers
- using factors to find out LCM of numbers

Attitudes and values

Ensure that the learners appreciate the importance of highest common factors and lowest common multiples.

Special needs education

Ensure that the activities are adapted for learners with special educational needs. This will help them develop their potentials and become independent citizens. Observe and record their performance.

Background information

Highest common factors (HCF) and Lowest Common Multiples (LCM) are often confused by learners. It is important to give learners adequate practice of HCF before starting LCM.

Pay special attention to the use of continued division in finding HCF, as most learners find this difficult. Lead them to realize that from the second stage, onwards it is the **remainder** that becomes the divisor. The last divisor becomes t HCF when division stops. For example the HCF of 120 and 400 can be calculated as follows:

a

	step 1		
3	120	400	3
	– 120	–360	
		step 2	
	0	40	

400 ÷ 120 = 3 remainder 40
Then 120 ÷ 40 = 3 remainder
Division stops when the remaind
becomes zero. The last divisor (4
becomes the HCF.

Similarly, the HCF of 480 and 1060 can be calculated as follows:

b

	step 1		
4	480	1060	2
	– 400	–960	
	step 3	step 2	
4	80	100	1
	80	– 80	
		step 4	
	0	20	

1,060 ÷ 480 = 2 remainder 1
480 ÷ 100 = 4 remainder 80
100 ÷ 80 = 1 remainder 2
80 ÷ 20 = 4 remainder 0
Division stops since the remaind
is zero at step 4 . The last divisor
(20) becomes the HCF.

Ensure that learners are given enough time to practice finding HCF using this method.

Activity 1 Finding HCF of numbers using factor method

Time allocation 2 lessons

Suggested teaching, learning and assessment resources

Multiplication tables, checklist, braille, mirror, feathers.

Instructions

1. Let learners be in groups
2. Ask learners to express 14, 4 and 26 as products of their prime factors
3. Let learners identify factors that occur to all the numbers
4. Ask learners the name given to such factors that occur to a set of given numbers (common factors)
5. Discuss with learners how to find the HCF of 14, 4 and 26 as follows:

   $14 = 2 \times 7$

   $4 = 2 \times 2$

   $26 = 2 \times 13$

   Common factor is, 2

   $\therefore$ HCF = 2
6. Let learners do exercise 5A from the learners book.

Activity 2 Finding HCF of numbers using division method

Time allocation 2 lessons

Suggested teaching, learning and assessment resources

Multiplication tables, checklist, braille, mirror, feathers.

Instructions

1. Let learners be in groups
2. Ask learners to find the HCF of 18, 24 and 48 using division method
3. Let learners explain how they arrived at the solution

4 Discuss with learners the working sequence as follows:

2	18	24	48
2	9	12	24
2	9	6	12
2	9	3	6
3	9	3	3
3	3	1	1
	1	1	1

HCF = 2 x 3 = 6

5 Let learners do exercise 5B from the learners book.

Activity 3 Finding HCF of numbers using continued division

Time allocation 2 lessons

Suggested teaching, learning and assessment resources

Multiplication tables, checklist, braille, mirror, feathers.

Instructions

1 Let learners be in groups
2 Ask learners to find the HCF of 480 and 1,060 by continued division
3 Let learners explain how they solved the problem
4 Discuss with learners the working sequence as follows:

4	480	1060	2
	400	960	
4	80	100	1
	80	80	
	0	20	

$\therefore$ HCF = 20

5 Discuss the examples in the learners' book.
6 Let learners do Exercise 5C from the learners book

Activity 4 Finding the LCM of numbers up to four terms using factor method

Time allocation 2 lessons

Suggested teaching, learning and assessment resources

Multiplication tables, division chart, braille, mirror, feathers.

Instructions

1 Let learners be in groups

2 Ask learners to find the prime factors of 40, 32, 24 and 16

3 Let them express each number as a product of its prime factors as shown below:

$40 = 2 \times 2 \times 2 \times 5$

$32 = 2 \times 2 \times 2 \times 2 \times 2$

$24 = 2 \times 2 \times 2 \times 3$

$16 = 2 \times 2 \times 2 \times 2$

The common prime factors are 2, 2 and 2.

Non common prime factors are 2, 2, 3 and 5

4 Discuss with learners that LCM of 40, 32, 24 and 16 can be calculated using the common and non common prime factors as follows:

$\therefore$ LCM = $2 \times 2 \times 2 \times 2 \times 2 \times 3 \times 5 = 480$

5 Help learners to find the LCM of the same numbers by dividing the numbers by their prime factors as follows:

2	40	32	24	16
2	20	16	12	8
2	10	8	6	4
2	5	4	3	2
2	5	2	3	1
3	5	1	1	1
5	1	1	1	1

$\therefore$ LCM = $2 \times 2 \times 2 \times 2 \times 2 \times 3 \times 5 = 480$

6 Discuss the example in the learners' book.

7 Let learners do exercise 5D from the learners book.

Activity 5 Solving practical problems on HCF and LCM

Time allocation 2 lessons

Suggested teaching, learning and assessment resources

Multiplication tables, division chart, checklist, braille, mirror, feathers.

Instructions

1 Discuss with learners the examples below:

a. What is the biggest number of children that would share equally, 40 mangoes, 16, bananas and 30 oranges without a remainder?

$40 = 2 \times 2 \times 2 \times 5$

$16 = 2 \times 2 \times 2 \times 2$

$30 = 2 \times 3 \times 5$

$HCF = 2$

$\therefore$ the number of children is 2

b. What is the smallest number of groundnuts that can be put into sachets of 20, 32, 24 and 40 without a remainder.

$20 = 2 \times 2 \times 5$

$32 = 2 \times 2 \times 2 \times 2 \times 2$

$24 = 2 \times 2 \times 2 \times 3$

$40 = 2 \times 2 \times 2 \times 5$

$LCM = 2 \times 2 \times 2 \times 2 \times 2 \times 3 \times 5 = 480$

$\therefore$ The number of groundnuts is 480

2 Let learners do exercise 5E from the learners book

Assessment

Use the table below to assess each learner.

Name of the learner: ____________________

How best is the learner able to:	Excellent	Good	Average	Needs support
1 find HCF of numbers using factor method?				
2 find HCF of numbers using division methods?				
3 find LCM of numbers up to 4 terms?				
4 find LCM of numbers using factor method?				

UNIT 4 Basic operations on fractions

Time allocation 19 lessons

Introduction

In everyday life situations, learners work with fractions in buying and selling items. They also use fractions in sharing items such as food, money and portions of space when playing. In this unit, learners will be carrying out any two basic operations on fractions in the same problem. They will also solve practical problems involving any two basic operations on fractions in the same problem.

Success criteria

Learners must be able to carry out any two basic operations on fractions in the same problem.

Developmental areas

Skills

Ensure that the learners develop the following skills:

- adding and subtracting fractions in the same problem
- adding and multiplying fractions in the same problem
- adding and dividing fractions in the same problem
- subtracting and dividing fractions in the same problem
- subtracting and multiplying fractions in the same problem
- multiplying and dividing fractions

Concepts and knowledge

Ensure that the learners understand and acquire the concepts and knowledge of basic operations on fractions.

Attitudes and values

Ensure that the learners appreciate the importance of basic operations on fractions.

Special needs education

Ensure that the activities are adapted for learners with special educational needs. This will help them develop their potentials and become independent citizens. Observe and record their performance.

Background information

In unit 3 learners solved problems on whole numbers that involve two or more basic operations. They used the recommended order of carrying out basic operations.

This knowledge will help them solve problems on fractions that involve more than one basic operation. They will also require the knowledge of LCM which they learnt in unit 5.

Ensure that learners are provided with enough support when they are carrying out subtraction of mixed numbers that involve regrouping.

For example

$$5\frac{1}{5} - 2\frac{7}{10} + 3\frac{3}{10}$$

$$= 5\frac{1}{5} + 3\frac{3}{10} - 2\frac{7}{10}$$

$$= 8\frac{2+3}{10} - 2\frac{7}{10}$$

$$= 8\frac{5}{10} - 2\frac{7}{10}$$

$$= 6\frac{5-7}{10}$$

Since 7 cannot be subtracted from 5, we take 1 from the whole number 6 and change it to $\frac{10}{10}$, then add it to $\frac{5}{10}$ to get $\frac{15}{10}$. The process continues as follows:

$$= 6\frac{5-7}{10} \text{ becomes } 5\frac{15-7}{10}$$

$$= 5\frac{8}{10}$$

$$= 5\frac{4}{5}$$

Activity 1 Adding and subtracting proper fractions

Time allocation 2 lessons

Suggested teaching, learning and assessment resources

Checklist, fraction chart, learners' experiences, braille, mirror, feathers.

Instructions

1 Ask learners to be in groups.

2 Let learners work out the following $^{2}/_{5} + {}^{3}/_{4} - {}^{7}/_{10}$.

3 Ask learners to report how they have come up with the answer.

4 Discuss with learners the working process as follows:

$$\frac{2}{5} + \frac{3}{4} - \frac{7}{10}$$

$$= \frac{8 + 15 - 14}{20}$$

$$= \frac{23 - 14}{20}$$

$$= \frac{9}{20}$$

5 Let the learners do exercise 6A from the learners' book.

Activity 2 Adding and subtracting mixed fractions

Time allocation 2 lessons

Suggested teaching, learning and assessment resources

Learners' experiences, checklist, braille, mirror, feathers.

Instructions

1 Let learners be in groups.

2 Ask learners to solve the following:(a) $7\,{}^{4}/_{5} + 6\,{}^{5}/_{6} - 11\,{}^{1}/_{4}$ (b) $5\,{}^{1}/_{5} - 2\,{}^{7}/_{10} + 3\,{}^{3}/$

3 Ask them to report their working sequence

4 Discuss with learners how to solve the problems as follows:

a. $7\frac{4}{5}+6\frac{5}{6}-11\frac{1}{4}$

$=(7+6)\frac{24+25}{30}-11\frac{1}{4}$

$=13\frac{49}{30}-11\frac{1}{4}$

$=14\frac{19}{30}-11\frac{1}{4}$

$=3\frac{38\text{-}15}{60}$

$=3\frac{23}{60}$

b. $5\frac{1}{5}-2\frac{7}{10}+3\frac{3}{10}$

$=5\frac{1}{5}+3\frac{3}{10}-2\frac{7}{10}$

$=8\frac{2+3}{10}-2\frac{7}{10}$

$=8\frac{5}{10}-2\frac{7}{10}$

$=6\frac{5\text{-}7}{10}$

$=5\frac{15\text{-}7}{10}$

$=5\frac{8}{10}$

5 Let the learners do exercise 6B from the learners' book.

Activity 3 Multiplying and dividing fractions

Time allocation 3 lessons

Suggested teaching, learning and assessment resource

Problem cards, learners' experience, braille, mirror, feathers.

Instructions

1 Ask the learners to be in groups.

2 Let learners work out the following: **a.** ${}^{7}/_{12} \div {}^{20}/_{21} \times {}^{8}/_{11}$ **b.** ${}^{3}/_{5} \div 7\,{}^{1}/_{2} \times 3\,{}^{3}/_{4}$

3 Ask learners to report their work.

4 Discuss with the learners how to solve these problems as follows:

a.
$$\frac{7}{12}\div\frac{20}{21}\times\frac{8}{11}$$
$$=\frac{7}{\cancel{12}_{\cancel{3}_1}}\times\frac{\cancel{21}^{7}}{\cancel{20}_{10}}\times\frac{\cancel{8}^{\cancel{2}^{1}}}{11}$$
$$=\frac{49}{110}$$

b.
$$\frac{3}{5}\div 7\frac{1}{2}\times 3\frac{3}{4}$$
$$=\frac{3}{5}\div\frac{15}{2}\times\frac{15}{4}$$
$$=\frac{3}{5}\times\frac{\cancel{2}^{1}}{\cancel{15}_{1}}\times\frac{\cancel{15}^{1}}{\cancel{4}_{2}}$$
$$=\frac{3}{10}$$

5 Discuss the examples in the learners book.

6 Let the learners do exercise 6C from the learners' book.

Activity 4 Adding and multiplying fractions

Time allocation 2 lessons

Suggested teaching, learning and assessment resources

Problem cards, checklist, learners' experiences, braille, mirror, feathers.

Instructions

1 Let learners be in groups.

2 Let them work out the following in their groups

a $\;{}^{2}/_{5} + {}^{7}/_{9} \times {}^{9}/_{10}$ b $\;(1\,{}^{1}/_{2} + 2\,{}^{1}/_{3}) \times 1\,{}^{1}/_{5}$

3 Let the learners present their work

4 Discuss with learners the working sequence as follows:

a. $\frac{2}{5}+\frac{7}{9}\times\frac{9}{10}$

$=\frac{2}{5}+\left(\frac{7}{\cancel{9}_1}\times\frac{\cancel{9}^1}{10}\right)$

$=\frac{2}{5}+\frac{7}{10}$

$=\frac{4+7}{10}$

$=\frac{11}{10}$

$=1\frac{1}{10}$

b. $\left(1\frac{1}{2}+2\frac{1}{3}\right)\times1\frac{1}{5}$

$=\left(3\frac{3+2}{6}\right)\times1\frac{1}{5}$

$=3\frac{5}{6}\times1\frac{1}{5}$

$=\frac{23}{\cancel{6}_1}\times\frac{\cancel{6}^1}{5}$

$=\frac{23}{5}$

$=4\frac{3}{5}$

5 Discuss the examples in the learners' book.

6 Let the learners do exercise 6D from the learners' book.

Activity 5 Adding and dividing fractions

Time allocation 3 lessons

Suggested teaching, learning and assessment resources

Problem cards, learners' experiences, braille, mirror, feathers.

Instructions

1 Let learners be in groups.

2 Ask learners to work out the following:

a $\frac{2}{15}+\frac{4}{9}\div\frac{2}{3}$

b $7\frac{1}{3}\div1\frac{5}{9}+8\frac{3}{14}$

3 Let learners report their work.

4 Discuss with learners the working sequence as follows:

a. $\frac{2}{15}+\frac{4}{9}\div\frac{2}{3}$

$=\frac{2}{15}+\left(\frac{4}{9}\div\frac{2}{3}\right)$

$=\frac{2}{15}+\left(\frac{\cancel{4}^{2}}{\cancel{9}_{3}}\times\frac{\cancel{3}^{1}}{\cancel{2}_{1}}\right)$

$=\frac{2}{15}+\frac{2}{3}$

$=\frac{2+10}{15}$

$=\frac{\cancel{12}^{4}}{\cancel{15}_{5}}$

$=\frac{4}{5}$

b. $7\frac{1}{3}\div1\frac{5}{9}+8\frac{3}{14}$

$=7\frac{1}{3}\div1\frac{5}{9}+8\frac{3}{14}$

$=\left(7\frac{1}{3}\div1\frac{5}{9}\right)+8\frac{3}{14}$

$=\left(\frac{22}{3}\div\frac{14}{9}\right)+8\frac{3}{14}$

$=\left(\frac{\cancel{22}^{11}}{\cancel{3}_{1}}\times\frac{\cancel{9}^{3}}{\cancel{14}_{7}}\right)+8\frac{3}{14}$

$=\frac{33}{7}+8\frac{3}{14}$

$=4\frac{5}{7}+8\frac{3}{14}$

$=12\frac{10+3}{14}$

$=12\frac{13}{14}$

5 Discuss the examples in the learners' book.

6 Let the learners do exercise 6E from the learners' book.

Activity 6 Subtracting and dividing fractions

Time allocation 2 lessons

Suggested teaching, learning and assessment resources

Learners' experiences, problem cards, braille, mirror, feathers.

Instructions

1 Ask the learners to be in groups.

2 Let the learners solve the following problems.

a $1\frac{1}{7}+\frac{3}{14}-\frac{4}{5}$ b $\left(\frac{3}{4}-\frac{3}{8}\right)+\frac{5}{6}$

3 Let the learners report their work.

4 Discuss with learners how to solve the problem as follows:

a. $1\frac{1}{7} \div \frac{3}{14} - \frac{4}{5}$

$= \left(1\frac{1}{7} \div \frac{3}{14}\right) - \frac{4}{5}$

$= \left(\frac{8}{7} \div \frac{3}{14}\right) - \frac{4}{5}$

$= \left(\frac{8}{\cancel{7}_1} \times \frac{\cancel{14}^2}{3}\right) - \frac{4}{5}$

$= \frac{16}{3} - \frac{4}{5}$

$= 5\frac{1}{3} - \frac{4}{5}$

$= 5\frac{5\text{-}12}{15}$

$= 4\frac{20\text{-}12}{15}$

$= 4\frac{8}{15}$

b. $\left(\frac{3}{4} - \frac{3}{8}\right) \div \frac{5}{6}$

$= \frac{6-3}{8} \div \frac{5}{6}$

$= \frac{3}{\cancel{8}_4} \times \frac{\cancel{6}^3}{5}$

$= \frac{9}{20}$

5 Discuss the examples in the learners' book.

6 Let the learners do exercise 6F from the learners' book.

Activity 7 Subtracting and multiplying proper fractions

Time allocation 2 lessons

Suggested teaching, learning and assessment resources

Problem cards, learners' experiences, braille,mirror, feathers.

Instructions

1 Let learners be in groups.

2 Ask learners to work out the following: $^2/_3 - {}^1/_4 \times {}^1/_3$

3 Let learners report their working procedure

4 Discuss with learners the working sequence as follows:

a. $\frac{2}{3}-\frac{1}{4}\times\frac{1}{3}$

$=\frac{2}{3}-\left(\frac{1}{4}\times\frac{1}{3}\right)$

$=\frac{2}{3}-\frac{1}{12}$

$=\frac{8-1}{12}$

$=\frac{7}{12}$

b. $\left(4\frac{1}{4}-2\frac{1}{2}\right)\times3\frac{1}{5}$

$=\left(2\frac{1-2}{4}\right)\times3\frac{1}{5}$

$=\left(1\frac{5-2}{4}\right)\times3\frac{1}{5}$

$=1\frac{3}{4}\times3\frac{1}{5}$

$=\frac{7}{\cancel{4}_1}\times\frac{\cancel{16}^4}{5}$

$=\frac{28}{5}$

$=5\frac{3}{5}$

5 Let the learners do exercise 6G from the learners' book.

Activity 8 Solving practical problems involving basic operations on fractions

Time allocation 2 lessons

Suggested teaching, learning and assessment resources

Checklist, learners' experiences, braille,mirror, feathers.

Instructions

1 Let learners be in groups.
2 Ask learners to solve the following:
From $^9/_{10}$ subtract the product of $^3/_4$ and $^2/_3$.
3 Let learners report their work.

4 Discuss with learners the working sequence as follows:

$$\frac{9}{10}-\frac{3}{4}\times\frac{2}{3}$$

$$=\frac{9}{10}-\left(\frac{\cancel{3}^1}{\cancel{4}_2}\times\frac{\cancel{2}^1}{\cancel{3}_1}\right)$$

$$=\frac{9}{10}-\frac{1}{2}$$

$$=\frac{9-5}{10}$$

$$=\frac{4}{10}$$

$$=\frac{2}{5}$$

5 Discuss the example in the learners' book.

6 Let the learners do exercise 6H from the learners' book.

Review exercise

Time allocation 1 lesson

Ask learners to do the review exercise at the end of unit 6.

Assessment

Use the table below to assess each learner.

Name of the learners: ____________________

How best is the learner able to:	Excellent	Good	Average	Needs support
1 add and subtract fractions in the same problem?				
2 multiply and divide fractions in the same problem?				
3 add and multiply fractions in the same problem?				
4 add and divide fractions in the same problem?				
5 subtract and divide fractions in the same problem?				
6 subtract and multiply fractions in the same problem?				

UNIT 7 Basic operations on decimals

Time allocation 20 lessons

Introduction

In the previous classes, learners were carrying out the basic operations on decimals separately. However, in real life situations, there are cases where learners may need to use two or more basic operations on decimals at the same time. Therefore it is necessary to equip learners with the skills of handling several operations in the same problem.

In this unit, learners will be carrying out two or more operations in the same problem. They will also be solving practical problems involving two or more basic operations on decimals.

Success criteria

Learners must be able to carry out any two basic operations on decimal numbers in the same problem.

Developmental areas

Skills

Ensure that the learners acquire the following skills:

- adding and subtracting decimals in the same problem
- multiplying and dividing decimals in the same problem
- adding and multiplying decimals in the same problem
- adding and dividing decimals in the same problem
- subtracting and dividing decimals in the same problem
- subtracting and multiplying decimals in the same problem
- solving practical problems involving basic operations on decimals

Concept and knowledge

Ensure that learners acquire the concepts and knowledge of basic operations on decimals.

Attitudes and values

Ensure that learners appreciate the importance of basic operations on decimals.

Special needs education

Ensure that the activities are adapted for learners with special educational needs. This will help them develop their potentials and become independent citizens. Observe and record their performance.

Background information

Basic operations on decimals are carried out in the same way as basic operations on whole numbers. The only difference is the presence of decimal point in decimal numbers. Learners are therefore expected to apply the knowledge which they acquired in unit 3 when carrying out basic operation on decimals.

When multiplying and dividing decimals most learners find it difficult to handle the decimal point. Therefore ensure that the learners are given adequate support.

Activity 1 Adding and subtracting decimals

Time allocation 2 lessons

Suggested teaching and learning resources

Place value chart, learners' experiences, braille, mirror, feathers.

Instructions

1 Let the learners be in groups
2 Ask them to identify the place values of digits in the following decimals: 0.19, 0.2, 0.01
3 Ask learners to work out the following problem: 0.19 + 0.2 – 0.01
4 Let learners present their work.
5 Discuss with learners to establish the answer as follows: 0.19 + 0.2 – 0.01

$$\begin{array}{r} 0.19 \\ +\ 0.2 \\ \hline 0.39 \\ -\ 0.01 \\ \hline 0.38 \\ \hline \end{array}$$

6 Let learners do exercises 7A from the learners' book.

Activity 2 Multiplying and dividing decimals

Time allocation 3 lessons

Suggested teaching and learning resources

Place value chart, learners' experiences, checklist, braille, mirror, feathers.

Instructions

1 Let learners be in groups
2 Ask learners to solve the following: 0.24 x 1.6 ÷ 0.8
3 Let learners report their work

4 Discuss with learners to establish the working sequence as follows:

$0.24 \times 1.6 \div 0.8$

$= 0.24 \times (1.6 \times 10 \div 0.8 \times 10)$

$= \dfrac{0.24 \times \cancel{16}^{2}}{\cancel{8}_{1}}$

$= 0.24 \times 2$

$= 0.48$

5 Discuss the example in the learners' book.

6 Let the learners do exercise 7B from the learners' book.

Activity 3 Addition and multiplication of decimals

Time allocation 3 lessons

Suggested teaching, learning and assessment resources

Learners' experiences, checklist, braille, mirror, feathers.

Instructions

1 Let the learners be in groups.

2 Ask learners to work out the following: $12.6 + 49.5 \times 2.6$

3 Let each group report their work.

4 Discuss with the learners to establish the working sequence as follows:

$12.6 + 49.5 \times 2.6$

$12.6 + (49.5 \times 2.6)$

$$\begin{array}{r} 49.5 \\ \times\ 2.6 \\ \hline 2970 \\ +\ 990 \\ \hline 128.70 \\ \hline \end{array}$$

$12.6 + 128.70$

$= 141.3$

5 Discuss the following example with learners:

$$9.5 \times (3.62 + 5.4)$$

$$
\begin{array}{lrrrrr}
= & & & 3. & 6 & 2 \\
 & & + & 5. & 4 & \\
\hline
 & & & 9. & 0 & 2 \\
 & & \times & 9. & 5 & \\
\hline
 & & 4 & 5 & 1 & 0 \\
+ & 8 & 1 & 1 & 8 & \\
\hline
 & 8 & 5. & 6 & 9 & 0 \\
\hline
\end{array}
$$

6 Discuss the examples in the learners' book.

7 Let the learners do exercise 7C from the learners' book.

Activity 4 Addition and division of decimals

Time allocation 2 lessons

Suggested teaching, learning and assessment resources

Place value chart, learners' experiences, checklist, braille, mirror, feathers.

Instructions

1 Let learners be in groups.

2 Let learners work out the following: $11.5 + 104.55 \div 12.3$

3 Let the groups present their work.

4 Discuss with learners the working sequence as follows:

$$
\begin{aligned}
& 11.5 + 104.55 \div 12.3 \\
&= 11.5 + (104.55 \div 12.3) \\
&= 11.5 + \frac{(1045.5)}{123} \\
&= 11.5 + 8.5 \\
&= 20.0
\end{aligned}
$$

5 Discuss the examples in the learners' book.

6 Let the learners do exercise 7D from the learners' book.

Activity 5 Subtraction and division of decimals

Time allocation 3 lessons

Suggested teaching, learning and assessment resources

Learners' experiences, checklist, braille, mirror, feathers.

Instructions

1 Let learners be in groups.

2 Ask learners to solve the following: $36.15 - 56.7 \div 5.4$

3 Let learners present their work.

4 Discuss with learners the working sequence as follows:

$136.15 - 56.7 \div 5.4$

$= 36.15 - (56.7 \div 5.4)$

$= 36.15 - 10.5$

$= 25.65$

5 Let the learners do exercise 7E from the learners' book.

Activity 6 Subtraction and multiplication of decimals

Time allocation 3 lessons

Suggested teaching, learning and assessment resources

Place value chart, learners' experiences, checklist, braille, mirror, feathers.

Instructions

1 Let learners be in groups.

2 Ask learners to work out the following: $5.0 \times 0.7 - 2.3$

3 Let learners report their work.

4 Discuss with learners to establish the working sequence as follows:

$5.6 \times 0.7 - 2.3$

$= (5.6 \times 0.7) - 2.3$

$$\begin{array}{r} 5.6 \\ \times\ 0.7 \\ \hline 3.92 \\ \hline \end{array}$$

$3.92 - 2.3$

$$\begin{array}{r} 3.92 \\ -\ 2.30 \\ \hline 1.62 \\ \hline \end{array}$$

5 Discuss the example in the learners' book.

6 Let the learners do exercise 7F from the learners' book.

Activity 7 Solving practical problems involving basic operations on decimals

Time allocation 3 lessons

Suggested teaching, learning and assessment resources

Learners' expereinces, braille, mirror, feathers, problem sheets.

Instructions

1 Ask learners to be in groups
2 Let learners work out the following: Divide 0.729 by 0.09 and multiply the result by 4.5.
3 Let learners report their work.
4 Discuss with learners to establish the working sequence as follows:

$$
\begin{aligned}
& 0.729 \div 0.09 \times 4.5 \\
= \; & (0.729 \times 100) \div (0.09 \times 100) \times 4.5 \\
= \; & 7.29 \div 9 \times 4.5 \\
& 0.81 \\
= \; & \frac{\cancel{7.29}^{0.81}}{\cancel{9}_{1}} \times 4.5 \\
= \; & 0.81 \times 4.5
\end{aligned}
$$

$$
= \quad \begin{array}{r}
0.81 \\
\times \quad 4.5 \\
\hline
4\;0\;5 \\
+ \quad 3\;2\;4 \\
\hline
3.\,6\;4\;5 \\
\hline
\end{array}
$$

$$\therefore \quad 0.729 \div 0.09 \times 4.5 = 3.645$$

5 Discuss the example in the learners' book.
6 Let the learners do exercise 7G from the learner's book

Review exercise

Time allocation 1 lesson

Ask learners to do the review exercise on unit 7.

Assessment

Use the table below to assess each learner.

Name of the learner: ______________________________

How best is the learner able to:	Excellent	Good	Average	Needs support
1 add and subtract decimals in the same problem?				
2 multiply and divide decimals in the same problem?				
3 add and multiply decimals in the same problem?				
4 add and divide decimals in the same problem?				
5 subtract and divide decimals in the same problem?				
6 subtract and multiply decimals in the same problem?				

Approximation and estimation

Time allocation 7 lessons

Introduction

In mathematics, accuracy is very important. Learners must develop the skill of accuracy in calculations, construction of angles and figures and drawing graphs. However, for them to reach this stage, they need to acquire the skill of estimating and predicting. Before coming up with the accurate answer, they should be able to estimate and predict the answer. After calculations they can compare their estimates with the answer. In this unit, learners will express numbers to a required degree of accuracy.

Success criteria

Learners must be able to express numbers to the required degree of accuracy.

Developmental areas

Skills

Ensure that the learners develop the following skills:

- writing decimal numbers up to 4 decimal places
- changing decimal numbers to mixed numbers and vice versa
- expressing numbers up to 3 significant figures

Knowledge and concepts

Ensure that the learners understand and acquire the following knowledge and concepts:

- approximation
- estimation
- significant figures

Attitudes and values

Ensure that the learners appreciate the importance of approximation and estimation in everyday life.

Special needs education

Ensure that the activities are adapted for learners with special educational needs. This will help them develop their potentials and become independent citizens. Observe and record their performance.

Background information

By now learners are able to carry out basic operations on numbers and decimals. They are also able to round up decimals to the nearest whole number and up to two decimal places. It is therefore important for learners to acquire the skill of rounding up numbers to the required degree of accuracy.

Significant figures on the other hand, are numbers that must be retained whatever the position of the decimal point. All the digits or figures in the number are important or significant.

Activity 1 Expressing decimal numbers up to 4 decimal places

Time allocation 2 lessons

Suggested teaching, learning and assessment resources

Place value chart, checklist, braille, mirror, feathers.

Instructions

1 Let learners be in groups.
2 Ask learners to explain the value of each of the following units on the place value chart.
  Th H T O t h th tth
3 Ask learners to give examples of numbers that will fit in the above units eg 4213.62733.
4 Discuss with learners how to round up numbers:
   a. When rounding up a number to four decimal places focus on the fourth and fifth digit after the decimal point.
   b. Round up the digit after the decimal point by 1 if the fifth digit is 5 or more, ie, 0.66666 becomes 0.6667 to 4 decimal places.
   c. Let the fourth digit after the decimal point remain the same if the fifth digit is less than 5, ie 0.84931 becomes 0.8493 to 4 decimal places.
5 Discuss the following examples with learners:
  write 8.436691 to 3 decimal places and 10.1123410 to 4 decimal places

  Answer

  a 8.436691 becomes 8.437 to 3 decimal places
  b 10.1123410 becomes 10.1123 to 4 decimal places
6 Discuss the example in the learners' book.
7 Let the learners do exercise 8A in the learners' book.

Activity 2 Changing decimal numbers to mixed numbers and vice versa

Time allocation 2 lessons

Suggested teaching, learning and assessment resources

A place value chart, braille, mirror, feathers.

Instructions

1 Let learners be in groups.
2 Ask learners to express the following as mixed numbers
   a 56.25 b 49.33
3 Let learners report their findings.
4 Discuss with learners the working sequence as follows:

a	56.25	b	49.33
=	$56^{25}/_{100}$	=	$49^{33}/_{100}$
=	$56^{1}/_{4}$	=	$49^{33}/_{100}$

5 Ask learners to express $3^{3}/_{7}$ as a decimal and write the answer to 2 decimal places.
6 Let learners report their work.
7 Discuss with learners to establish the working sequence as follows:

$$3\frac{3}{7} = \frac{24}{7}$$
$$= 3.428\ldots$$
$$= 3.43 \text{ to 2 decimal places}$$

8 Let learners do exercises 8B and 8C from learner's book.

Activity 3 Establishing the meaning of significant figures

Time allocation 2 lessons

Suggested teaching and learning resources

A place value chart, braille, mirror, feathers.

Instructions

1 Let learners be in groups.
2 Ask learners to discuss the place value of each digit in the following numbers: 67,384; 6,738.4; 673.84; 67.384; 6.7884
3 Let learners report their findings.
4 Help learners to establish that the digits are the same in each number but the numbers have different values because of the position of the decimal point.
5 Help learners to establish that each figure or digit is significant or important in the numbers.

6 Help learners to establish that significant figures are numbers that are retained whatever the position of the decimal point: eg 0.4352 becomes 0.435 to 3 significant figures.

Activity 4 Expressing numbers up to 3 significant figures

Time allocation 1 lesson

Suggested teaching, learning and assessment resources

Learners' experiences, checklist, braille, mirror, feathers.

Instructions

1 Let learners be in groups.

2 Discuss with learners the following examples:

Express the following numbers to the given number of significant figures

a 9218 to 2 significant figures b 0.36543 to 3 significant figures

c 0.040865 to 3 significant figures

Answer

a 9218 becomes 9200 to 2 significant figures

b 0.36543 becomes 0.365 to 3 significant figures

c 0.040865 becomes 0.0409 to 3 significant figures

3 Let the learners do exercise 8D in the learners' book.

Assessment

Use the table below to assess each learner.

Name of the learner: ______________________________

How best is the learner able to:	Excellent	Good	Average	Needs support
1 write decimal numbers up to 4 decimal places?				
2 change decimal numbers to mixed numbers?				
3 change mixed numbers to decimal numbers?				
4 express numbers up to 3 significant figures?				

UNIT 9 Rate, ratio and proportion

Time allocation 10 lessons

Introduction

Rate, ratio and proportion are related concepts that form our everyday experiences. The concepts of rate and ratio are not new to learners. In Standard 6, learners calculated the ratio of quantities and the rates of activities. In this unit learners will continue to calculating ratio, rate and proportion. They will also solve practical problems involving ratio, rate and proportion.

Success criteria

Learners must be able to:

- find rate of different activities
- find ratio of given quantities

Developmental areas

Skills

Ensure that learners acquire the following skills:

- finding rate of different activities
- finding ratio of given quantities
- solving practical problems involving ratio and proportion

Concepts and knowledge

Ensure that learners understand and acquire the following concepts and knowledge:

- rate
- ratio
- proportion

Attitudes and values

Ensure that the learners appreciate the importance of rate, ratio and proportion in everyday life.

Special needs education

Ensure that the activities are adapted for learners with special educational needs. This will help them develop their potentials and become independent citizens. Observe and record their performance.

Background information

Ratio is used when comparing two or more numbers or quantities. Ratios can express comparisons of a part to a whole eg the ratio of girls to all the learners in a class or part-to-part eg the ratio of number of boys to number of girls in class. A rate on the other hand is a comparison of the measures of two different things or quantities with different units. For example, speed is a rate that gives a measure of distance per unit time. A proportion is a statement of equality between two ratios. It describes how related any two ratios are.

Activity 1 Estimating speed from timed activities

Time allocation 2 lessons

Suggested teaching, learning and assessment resources
Stop watches, cups, sand, pails, learners' experiences, checklist, braille, mirror, feathers.

Instructions

1 Let learners be in groups.
2 Ask two learners to run a distance of 50m and record the time taken.
3 Ask learners to determine the speed of the two learners.
4 Discuss with learners how to work out speed using the following formula:

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}}$$

5 Let learners in their groups perform different activities to find speed eg running 100m race, 200m race.

Activity 2 Solving practical problems involving speed

Time allocation 2 lessons

Suggested teaching, learning and assessment resources
Learners' experience, braille, mirror, feathers.

Instructions

1 Let learners be in groups.
2 Ask learners to solve the following:
A lorry covers 265km in 2 hours. Find the speed of the lorry.
3 Ask learners to report their work.

4 Discuss with learners the working sequence for speed as follows:

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}}$$

$$= \frac{265 \text{ km}}{2 \text{ hr}}$$

$$= 132.5 \text{ km/hr}$$

5 Let learners do exercise 9A from the learners' book.

Activity 3 Simplifying ratios to their lowest terms

Time allocation 2 lessons

Suggested teaching, learning and assessment resources

Learners' experiences, checklist, braille, mirror, feathers.

Instructions

1 Ask learners to find the ratio for each of the following:
   a. a choir has 15 girls and 20 boys
   b. out of 800 learners at a certain school, 420 are girls
2 Help learners to establish that the ratios are:
   a 15 girls to 20 boys = 15:20
   b 800 learners to 420 girls = 800:420
3 Discuss with learners how to simplify these ratios to their lowest terms as follows:

$$15:20 = \frac{15}{20} = \frac{3}{4} = 3:4 \qquad 800:420 = \frac{800}{420} = \frac{40}{21} = 40:21$$

4 Help learners to understand that for every 3 girls in a choir there are 4 boys.
5 Discuss with learners the meaning of ratio.
6 Let learners do exercise 9B from the learners' book.

Activity 4 Sharing quantities in given ratios using unitary method

Time allocation 2 lessons

Suggested teaching, learning and assessment resources

Pens, books, money, stones, sticks checklist,, braille, mirror, feathers.

Instructions

1 Let learners be in groups.
2 Ensure that each group has 20 items of the same type.
3 Ask each group to share the items to 2 learners ie learner A and learner B in the ratio 2:3.
4 Let each group present their findings to the class.
5 Discuss with learners to establish that learner A gets 8 items while learner B gets 12 items.
6 Discuss the following example with learners:

   Mary and Thandie shared 126 oranges in the ratio 4:5. How many oranges did each one get?

   Answer

   Ratio 4:5

   Total parts of the ratio $= 4 + 5 = 9$

   One share $= \frac{126 \text{ oranges}}{9}$

   $= 14$ oranges

   Mary's share $= 14 \times 4 = 56$ oranges

   Thandie's share $= 14 \times 5 = 70$ oranges

7 Discuss the example in the learners' book.
8 Let learners do exercise 9C from the learners' book.

Activity 5 Using proportion in sharing quantities

Time allocation 1 lesson

Suggested teaching, learning and assessment resources

Pens, books, learners, checklist, braille, mirror, feathers.

Instructions

1 Ask two learners to share 12 books between them such that for every book that one learner gets the other one gets two books.

2 Let learners write the ratio of the number of books that each learner got ie 4:8 = 1:2

3 Ask learners to discuss the following example in groups:

Mrs Phiri shared 60 mangoes among her three children, Tawonga, Chimwemwe and Kondwani so that for every 1 mango Tawonga got, Chimwemwe got 2 and Wongani got 3. how many mangoes did each child get.

4 Let learners present their work.

5 Discuss with learners the working sequence as follows:

Total number of mangoes 60

Ratio of Tawonga : Chimwemwe : Kondwani = 1:2:3

Total parts of the ratio = 1 + 2 + 3 = 6

Taonga's share = $\frac{1}{6} \times 60$ mangoes = 10 mangoes

Chimwemwe's share = $\frac{2}{6} \times 60$ mangoes = 20 mangoes

Kondwani's share = $\frac{3}{6} \times 60$ mangoes = 30 mangoes

6 Discuss the example in the learners' book.

7 Let learner do exercise 9D from the learners' book.

Review exercise

Time allocation 1 lesson

Ask the learners to do the review exercise on unit 9.

Assessment

Use the table below to assess each learner.

Name of the learner: ______________________

How best is the learner able to:	Excellent	Good	Average	Needs support
1 estimate speed from timed activities?				
2 solve problems involving speed?				
3 simplify ratio to its lowest term?				
4 share quantities in given ratios using unitary method?				
5 use proportion in sharing quantitites?				

UNIT 10 Percentages

Time allocation 11 lessons

Introduction

The concept of percentages is used in everyday life. People use percentages in various ways such as in recording marks of learners and in representing population of a particular group of people. It is important that learners understand the concept as they will apply it in various situations.

In this unit, learners will be introduced to percentages, express percentages as fractions and vice versa, express percentages as decimals and vice versa and work out practical problems on percentages.

Success criteria

Learners must be able to work out percentages.

Developmental areas

Skills

Ensure that learners acquire the following skills:

- expressing percentages as fractions and vice versa
- expressing percentages as decimals and vice versa
- working out practical problems on percentages

Concept and knowledge

Ensure that learners understand and acquire the concept and knowledge of percentages

Attitudes and values

Ensure that the learners appreciate the importance of percentages.

Special needs education

Ensure that the activities are adapted for learners with special educational needs. This will help them develop their potentials and become independent citizens. Observe and record their performance.

Background information

By now learners have come across the word percentage for many times. Especially during assessment. They realise that by getting more correct answers, they are awarded a high percentage. Learners should understand percentage as out of hundred. Help learners realise that fractions whose denominator is not hundred can also be changed into percentages. In addition decimals can also be expressed as percentages. However, decimals with more than 2 decimal places should be changed to 2 decimal places before expressing them as percentages, eg 0.753 to 2 decimal places becomes 0.75. As a percentage it becomes 75%.

Activity 1 Introducing percentages

Time allocation 1 lesson

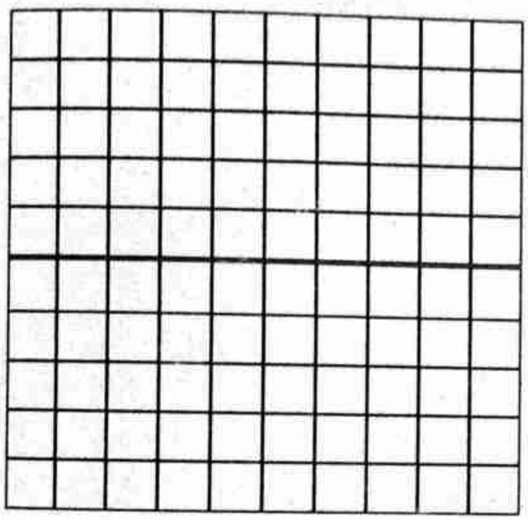
Suggested teaching, learning and assessment resources

100 square chart, pental pens, rulers, checklist, braille, mirror, feathers.

Instructions

1 Distribute paper and pencils to groups of learners.

2 Display 100 square chart like the one shown below:



3 Ask learners to shade on their charts the following:

a. 5 out of 100

b. 26 out of 100

c. 42 out of 100

4 Discuss with learners that percentage means out of one hundred.

5 Introduce to learners the symbol for percent as %.

6 Let learners write their answers as percentages.

Activity 2 Expressing percentages as fractions

Time allocation 2 lessons

Suggested teaching, learning and assessment resources

chart paper, pental pens, rulers, checklist, braille, mirror, feathers.

Instructions

1 Let learners be in groups.

2 Ask learners to shade on their square charts the following: 46%, 55% and 81%.

3 Let learners write down the fractions represented by the shaded parts.

4 Discuss with learners to establish that:

$$46\% = \frac{46}{100} = \frac{23}{50}$$

$$55\% = \frac{55}{100} = \frac{11}{20}$$

$$81\% = \frac{81}{100} \text{ expressed as fractions}$$

5 Discuss the example in the learners' book.

6 Let learners do exercise 10A from the learners' book.

Activity 3 Expressing fractions as percentages

Time allocation 2 lessons

Suggested teaching, learning and assessment resources

Paper, pencils, rulers, checklist, braille, raised pictures/diagrams, mirror, feathers.

Instructions

1 Let learners be in groups.

2 Ask learners to convert the following fractions into percentage:

$$\frac{25}{100}, \quad \frac{16}{20}, \quad \frac{1}{4}$$

3 Let learners present their work.

4 Discuss with learners the working sequence as follows:

$$\frac{25}{100} \text{ as percentage} = \frac{25}{100} \times 100\%$$
$$= 25\%$$

$$\frac{16}{20} \text{ as percentage} = \frac{16}{20} \times 100\%$$
$$= 80\%$$

$$\frac{1}{4} \text{ as a percentage} = \frac{1}{4} \times 100\%$$
$$= 25\%$$

5 Discuss the example in the learners' book.

6 Let learners do exercise 10B in the learners' book.

Activity 4 Expressing percentages as decimals

Time allocation 2 lessons

Suggested teaching, learning and assessment resources

Learners' experience, checklist, braille mirror, feathers.

Instructions

1 Let learners be in groups.
2 Ask learners to convert the following percentages into decimals: 1%, 34%, 60% and 112%.
3 Let learners report their work.
4 Discuss with learners the working sequence as follows:

$$1\% = \frac{1}{100} = 0.01$$

$$34\% = \frac{34}{100} = 0.34$$

$$60\% = \frac{60}{100} = 0.6$$

$$112\% = \frac{112}{100} = 1.12$$

5 Let learners do exercise 10C from the learners' book.

Activity 5 Expressing decimals as percentages

Time allocation 2 lessons

Suggested teaching, learning and assessment resources

Paper, pencils, rulers, checklist, braille, mirror, feathers.

Instructions

1 Let learners be in groups.
2 Ask learners to express the following decimals as percentages: 0.64, 0.03, 0.91, 0.7, 0.349.
3 Let learners present their work.

4 Discuss with learners the working sequence as follows:

0.64 as a percentage = $\frac{0.64}{100} \times 100 = \frac{64}{100} = 64\%$

0.03 as a percentage = $\frac{0.03}{100} \times 100 = \frac{3}{100} = 3\%$

0.91 as a percentage = $\frac{0.91}{100} \times 100 = \frac{91}{100} = 91\%$

0.7 as a percentage = $\frac{0.7}{100} \times 100 = \frac{70}{100} = 70\%$

0.349 as a percentage = $\frac{0.349}{100} \times 100 = \frac{35}{100} = 35\%$

5 Let learners do exercise 10D in the learners' book.

Activity 6 Solving practical problems involving percentages

Time allocation 1 lesson

Suggested teaching, learning and assessment resources

Learners' experience, braille, checklist, mirror, feathers.

Instructions

1 Let learners be in groups.
2 Ask learners to solve the following:
   a what is 25% of 300
   b in a class of 80 learners, 50 are girls. What is the percentage of girls in the class.
3 Let learners present their work.
4 Discuss the working sequence as follows:
   a. 25% of 300 = $\frac{25}{100} \times 300 = 3 \times 25 = 75$
   b. Percentage of girls in the class

$$\frac{\text{Numbers of girls}}{\text{Total number of learners}} \times 100\%$$

$$= \frac{50}{80} \times 100$$

$$= \frac{5}{8} \times 100\%$$

$$= 62.5\%$$

5 Discuss the example in the learners' book.
6 Let learners do exercise 10E from the learners' book.

Review exercise

Time allocation 1 lesson

Ask the learners to at the end of unit 10.

Assessment

Use the table below to assess each learner.

Name of the learner: ____

How best is the learner able to:	Excellent	Good	Average	Needs support
1 express percentages as fractions?				
2 express fractions as percentages?				
3 express percentages as decimals?				
4 express decimals as percentages?				

UNIT 11 Bills and budgets

Time allocation 5 lessons

Introduction

Learners buy different items for themselves and their families in everyday life. Some have been involved in budgeting. In this unit learners will prepare family budgets and budget for future activities such as parties and trips. They shall also work out bills.

Success criteria

Learners must be able to prepare budgets and work out bills

Developmental areas

Skills

Ensure that learners acquire the following skills:

- preparing budgets
- working out bills

Concepts and knowledge

Ensure that learners understand and acquire the following concepts and knowledge:

- budgets
- bills
- preparation of budgets and bills

Attitudes and values

Ensure that learners appreciate the importance of budgets and bills.

Special needs education

Ensure that the activities are adapted for learners with special educational needs. This will help them develop their potentials and become independent citizens. Observe and record their performance.

Background information

Learners have already been introduced to the preparation of budgets with items up to four.

The difference between a budget and a bill is that a budget is a general plan of items to be bought while a bill is an invoice that needs to be paid for services rendered or goods purchased. Some examples of common bills for family and institutions are: electricity, water, telephone and hospital bills. The bills come in as a result of services that have been provided and need payment.

Activity 1 Preparing budgets

Time allocation 2 lessons

Suggested teaching, learning and assessment resources

Shopping corner, price list chart, shopping items, price tags and local markets, checklist, braille, raised pictures/diagrams, mirror, feathers.

Instructions

1 Let learners be in groups.

2 Ask learners to work out the items they buy and their prices

3 Ask learners to prepare a budget for a number of items they have chosen using a price list chart for example:

2 pts of biscuit at K__________ each		= K__________
3 kg of meat at K__________/kg		= K__________
6 onions at K__________ each		= K__________
	total	= K__________

4 Discuss with learners the following example:
An entertainment committee came up with a list of items for a party: 2 chickens at K350 each, 5 heaps of tomatoes at K50 per heap, 5 bulbs of onions at K15 each, 12 packets of rice at K70 per packet. Write a budget for the committee.

– 2 chickens at K350 each	=	K 700.00
– 5 heaps of tomatoes at K50/ heap	=	K 250.00
– 7 bulbs of onion at K15 / bulb	=	K 105.00
– 12 packets of rice at K70/ packet	=	K 840.00
Total	=	K1,895.00

5 Discuss the example in the learners' book.

6 Let the learners do exercise 11A from the learners' book.

Activity 2 Working out bills

Time allocation 2 lessons

Suggested teaching, learning and assessment resources

Price list, checklist, braille, raised pictures/diagrams, mirror, feathers.

350
180
530

Instructions

1 Let learners be in groups.
2 Display a price list on the chalkboard or chart like the one shown below:

Price list

Soft drinks at K35 per bottle

A packet of salt at K20 per packet

Flour at K90 per kg

Cooking oil at K150 per *l*

3 Let learners work out the following using the price list.
Find the bill for Mrs Phiri who made the following budget:
7 litres of cooking oil; 10 bottles of soft drinks; 2 packets of salt; 5 kg flour.
4 Let learners present their work.
5 Discuss with learners the working sequence as follows:

7*l* cooking oil at K150 per litre	=	K1050.00
10 bottles of soft drinks at K35/bottle	=	K 350.00
2 packets of salt at K20 /packet	=	K 40.00
5 kg of flour at K90/kg	=	+ K 450.00
Total	=	K1890.00

6 Discuss the example in the learners' book.
7 Let learners do exercise 11B from the learners' book.

Review exercise

Time allocation 1 lesson

Let learners turn to the learners' book unit 11 and do the review exercise.

Assessment

Use the table below to assess each learner.

Name of the learner: ____________________

How best is the learner able to:	Excellent	Good	Average	Needs support
1 prepare budgets?				
2 work out bills?				

UNIT 12 Profit and loss

Time allocation 7 lessons

Introduction

Profit and loss are commonly used terms in business. People do business in order to make profits although sometimes they make losses. It is important to teach learners the concepts of profit and loss in order to equip them with business knowledge and skills. In this unit, learners will be introduced to the idea of profit and loss.

Success criteria

The learners must be able to find profit and loss.

Developmental areas

Skills

Ensure that the learners acquire the following skills:

- identifying cost and selling prices
- calculating profit and loss
- finding cost price given selling price and profit or loss
- finding selling price given cost price and profit or loss

Concepts and knowledge

Ensure that the learners understand and acquire the following concepts and knowledge:

- profit and loss
- cost price
- selling price

Attitudes and values

Ensure that the learners appreciate the importance of proper planning in order to maximise profit and avoid loss.

Special needs education

Ensure that the activities are adapted for learners with special educational needs. This will help them develop their potentials and become independent citizens. Observe and record their performance.

Background information

The ideas of profit and loss are not new to learners. Most of them have had experiences of deducing whether a profit or loss has been made. Therefore, in this unit the learners are building on their previous experiences. They will calculate profit and loss, find cost price given selling price and profit or loss and find selling price given cost price and profit or loss.

Activity 1 Identifying cost and selling prices

Time allocation 2 lessons

Suggested teaching, learning and assessment resources

Two shopping corners (wholesale, retail), learners experiences, checklist, braille, raised pictures/diagrams, mirror, feathers.

Instructions

1 Prepare two shopping corners, wholesale and retail with items priced as follows:

Item	Wholesale price	Retail
50g salt	K10.00	K12.00
washing soap	K28.00	K25.00
cooking oil	K40.00	K40.00
chitenje	K350.00	K400.00
biscuit	K20.00	K25.00
tinned fish	K40.00	K36.00

2 Ask learners to identify prices of items from the two shopping corners.
3 Discuss with learners why there is a difference in prices of the same item from the two shopping corners.
4 Help learners to understand that one is cost price and the other one is a selling price, some items will be sold at a profit, others will be sold at a loss.
5 Let learners practice buying from the wholesale and selling at retail prices.
6 Discuss the following example with learners:
a shopkeeper buys a bottle of cooking oil at K150.00 and sells it at K175.00. Identify cost and selling prices.
7 Let learners do exercise 12A from the learners' book.

Activity 2 Calculating profit and loss

Time allocation 2 lessons

Suggested teaching, learning and assessment resources

learners experiences, braille, mirror, feathers.

Instructions

1 Let learners be in groups.
2 Ask learners to calculate the profit and loss in each of the following:
   a. A bicycle was bought at K6,500.00 and sold at K6,090.00
   b. A farmer bought a piece of land at K22,000.00 and sold it at K36,000.00.

3 Discuss with learners to establish the working sequence as follows:

a. Profit = Selling price − Cost price
= K36,000.00 − K22,000.00
= K14,000.00

b. Loss = Cost price − Selling price
= K6,500.00 − K6,090.00
= K410.00

4 Discuss the examples in the learners' book.

5 Let learners do exercise 12B from the learners' book.

Activity 3 Finding cost price given selling price and profit or loss

Time allocation 1 lesson

Suggested teaching, learning and assessment resources

Learners experiences, checklist, braille, mirror, feathers.

Instructions

1 Let learners be in groups.

2 Ask learners to work out the following:

a. A chair was sold at K1,750.00 and at a profit of K450.00. Find the cost price.

b. A bicycle was sold at K6,500 at a loss of K300. Find the cost price.

3 Let learners present their work.

4 Discuss with learners the working sequence as follows:

a. Cost price = Selling price − profit
= K1,750.00 − K450.00
= K1,300.00

b. Cost price = Selling price + Loss
= K6,500.00 + K300.00
= K6,800.00

5 Discuss the example in the learners' book.

6 Let learners do exercise 12C from learners book.

Activity 4 Finding selling price given cost price and profit or loss

Time allocation 1 lesson

Suggested teaching, learning and assessment resources

Learners experiences, checklist, braille, mirror, feathers.

Instructions

1 Let learners be in groups.

2 Ask learners to work out the following:

   a. A hoe was bought at K350.00 and sold at a profit of K75.00. Find the selling price.

   b. A basket was sold at a loss of K80.00. If it was bought at K210.00, find the selling price.

3 Let learners present their work.

4 Discuss with learners the working sequence as follows:

a. Selling price	=	Cost price + profit
	=	K350.00 + K75.00
	=	K425.00
b. Selling price	=	Cost price – Loss
	=	K210.00 – K80.00
	=	K130.00

5 Discuss the example in the learners' book.

6 Let learners do exercise 12D from the learners' book.

Review exercise

Time allocation 1 lesson

Ask the learners to do the review exercise at the end of unit 12.

Assessment

Use the table below to assess each learner.

Name of the learner: ____________________

How best is the learner able to:	Excellent	Good	Average	Needs support
1 identify cost and selling price?				
2 calculate profit and loss?				
3 find cost price given selling price and profit or loss?				
4 find selling price given cost price and profit or loss?				

UNIT 13 Commission and discount

Time allocation 6 lessons

Introduction

Learners have been involved in buying and selling of items where a discount has been offered. They might have offered a discount on items they sold or might have been given a discount on items they bought. A discount is given for various reasons such as to promote sales and clear out stock.

In some cases, sales-persons are given commission. This is money given for a service provided.

In this unit learners will be introduced to commission and discount.

Success criteria

Learners must be able to calculate commission and discount

Developmental areas

Skills

Ensure that learners acquire the following skills:

- calculate commission
- calculate discount
- calculate the marked and selling price

Concepts and knowledge

Ensure that learners understand and acquire the following concepts and knowledge:

- commission
- discount

Attitudes and values

Ensure that learners appreciate the importance of commission and discount.

Special needs education

Ensure that the activities are adapted for learners with special educational needs. This will help them develop their potentials and become independent citizens. Observe and record their performance.

Background information

Discount is the money deducted from the price of an item. For instance, if the price of an item is K50 and is sold at K45, the discount is K5. Commission is the money paid for services rendered, or for every item sold. Newspaper sellers are paid money for every newspaper they sell as commission.

Commission is given as an incentive to a seller for selling an item. The more items one sells the higher the commission one gets.

Activity 1 Finding discount given marked price and selling price

Time allocation 2 lessons

Suggested teaching, learning and assessment resources

Learners' experiences, checklist, braille, mirror, feathers.

Instructions

1 Discuss with the learners if they have ever been offered a discount or they have ever offered a discount during buying or selling of the commodities (let learners say their experiences).
2 Ask learners to work out the following question in their groups:
A cell phone which was marked at K25,000 was sold for K23,500. what was the discount?
3 Let learners present their work.
4 Discuss with learners the working sequence as follows:

Discount	=	marked price – selling price
	=	K25,000 – K23,500
	=	K1,500

5 Let learners do exercise 13A from the learners book.

Activity 2 Calculating marked price and selling price

Time allocation 2 lessons

Suggested teaching, learning and assessment resources

Checklist, learners' experiences, braille, mirror, feathers.

Instructions

1 Let learners be in groups.
2 Ask learners to work out the following:
A dress was bought at K815.00. If a discount of K35.00 was given, what was the marked price?
3 Let learners present their work.
4 Discuss with learners the working sequence as follows:

Marked price	=	discount + selling price
	=	K35.00 + K815.00
	=	K850.00

5 Ask learners to also work out the following:
A radio was marked at K2,100.00. What was its selling price if a discount of K350 was given.
6 Let learners present their work.

7 Discuss the example in the learners' book.
8 Let the learners do exercise 13B from the learners' book.

Activity 3 Calculating commission

Time allocation 2 lessons

Suggested teaching, learning and assessment resources

Learners' experience, braille,mirror, feathers.

Instructions

1 Ask learners to brainstorm the meaning of commission.
2 Discuss with learners the meaning of commission as the amount of money paid for services offered or items sold.
3 Discuss with learners situations where a commission is given.
4 Ask learners to work out the following:
A newspaper seller gets a commission of K2.00 on every newspaper sold. If he sold 247 newspapers? How much commission did he receive?
5 Let learners present their work.
6 Discuss with learners the working sequence as follows:

commission for 1 newspaper	=	K2.00
commission for 247 newspapers	=	K2.00 x 247
	=	K494.00

7 Discuss the following example with learners:
A saleslady earns a commission of K75.00 on every K1,000.00 realized from the sales . How much will she earn if she sold items up to K16,000.00?
K1,000.00 earns K75.00 commission

K16,000.00 earns $= \dfrac{K16,000 \times 75}{1,000}$

$= K1,200.$

8 Discuss the examples in the learners' book.
9 Let learners do exercise 13C from the learners' book.

Assessment

Use the scale below to assess each learner.

Name of the learner: ____________________

How best is the learner able to:	Excellent	Good	Average	Needs support
1 calculate discount?				
2 calculate commission?				
3 calculate marketd price?				
4 calculate selling price?				

UNIT 14 Postal services

Time allocation 18 lessons

Introduction

The post office renders different services including postal and electronic mail transfer. Most learners have seen post offices in their local areas. Others have sent or received letters and money through the post offices. However some of them do not know that there are more services offered by post offices than mail and money delivery. It is, therefore important to teach learners all the services offered by post offices so that they (children) can use the postal facilities effectively. In this unit learners will calculate charges for sending money orders, postal orders, letters, parcels and telegrams.

Success criteria

Learners must be able to calculate charges for postal services.

Developmental areas

Skills

Ensure that learners acquire the following skills:

- completing money orders, postal orders and telegrams
- calculating charges for sending money orders, postal orders, letters, parcels and telegrams.
- calculating commission for money orders and postal orders.

Concepts and knowledge

Ensure that learners understand and acquire the following concepts and knowledge:

- postal services in Malawi
- postal orders, telegrams and money orders.
- charges and commission on sending of money and postal orders, letters and parcels

Attitudes and values

Ensure that learners appreciate the importance of postal services.

Special needs education

Ensure that the activities are adapted for learners with special educational needs. This will help them develop their potentials and become independent citizens. Observe and record their performance.

Background information

Postal services deal with delivery of items such as letters, money and parcels. The post office charges some fees for sending such items. For sending letters and parcels, the cost depends on the mass of the item while for sending money, the cost depends on the amount to be sent. And for sending messages, the cost depends on the number of words. Apart from these charges, the post office also charges the registration commission and telegram rates for sending items.

In this unit, learners will be filling in money and postal orders and telegrams. They will also calculate charges for sending money orders, postal orders, letters, parcels and telegrams.

Activity 1 Postal services

Time allocation 2 lessons

Suggested teaching, learning and assessment resources

Resource person, post office, chart with latest postal charges, braille, raised pictures, mirror, feathers.

Instructions

1. Let learners be in groups.
2. Ask them to discuss services provided by post office.
3. Let them report their work.
4. Let the resource person/teacher discuss with learners the postal services.
   a. Ordinary mail service - The delivery of item such as letters and parcels sent by Post.
   b. Postal orders service - Delivery of money sent by post in the form of postal orders.
   c. Money order services - Delivery of money sent by post in form of money orders.
   d. Registered mail services - Delivery of items such as letters and parcels sent as registered mail.
   e. Telegram - This is the delivery of message or money sent by telegraph.
   f. Express mail service – the delivery of letters parcels or money sent as "express mail"

Activity 2 Calculating charges for sending letters

Time allocation 3 lessons

Suggested teaching, learning and assessment resources

Stamps of various values, envelopes, braille, chart of charges, letters of different masses, newspaper and post cards, braille, raised pictures/diagrams mirror, feathers.

Instructions

1. Let learners be in groups.
2. Ask learners to explain the process followed when one is sending a letter or any other item to another person by post.

3 Discuss with learners the process as follows:
   a the letter is weighed and charged according to mass
4 Discuss with learners how to find the postal charge for a letter of mass 127g as follows:
   - ask learners to find out from table 14.1 the appropriate charge
   - assist learners to establish that the postal charge for the letters is K195.00.
5 Discuss the example in the learners' book.
6 Let learners do exercise 14A from the learners book.

Activity 3 Calculating charges for sending registered items

Time allocation 2 lessons

Suggested teaching, learning and assessment resources

Chart of charges for registered items, sample registered mails, braille, raised pictures, mirror, feathers.

Instructions

1 Let learners be in groups.
2 Display a chart of charges for registered items.
3 Let them use the chart to calculate charges for sending by post the following registered items: parcels weighing
   a. 100g
   b. 345g
   c. 3,000g
4 Ask them to report their work.
5 Discuss with lesrners how to find the charges as follows:

a. 100g

Postal stamps	=	K195
Registered fee	=	K 25
Total		K220

b. 345g

Postal stamps	=	K370
Registration fee	=	K 25
Total		K395

c. 3,000g

Postal stamps for up to 2000g	=	K1,030
Registration fee	=	K 25
Additional 1,000g	=	K 520
Registration fee	=	K 25
Total		K1,600

6 Discuss the example in the learners' book.
7 Let learners do exercise 14B from the learners book.

Activity 4 Calculating commission and total charges for buying and sending postal orders

Time allocation 2 lessons

Suggested teaching, learning and assessment resources

Sample of postal orders, chart showing postal orders and commission rates, braille, raised pictures, mirror, feathers.

Instructions

1. Let learners be in groups.
2. Discuss with learners how to fill in the postal order correctly.
3. Let learners discuss the following:

   Mrs Tembo sent K5,000.00 by postal order to her daughter. Find the total charges of sending the money.
4. Let learners present their work.
5. Discuss with learners the working sequence as follows:

   Amount of postal order = K5,000.00

   Commission = 10%

   $\therefore$ Commission = $\dfrac{10 \times 5000}{100}$

   = K500.00

   $\therefore$ total charges = amount + commission + postage registration

   = K5,000 + K500 + K40 + K25

   = K5,565.00
6. Discuss the examples in the learners' book.
7. Let learners do exercise 14C from the learners book.

Activity 5 Calculating commission and charges for buying and sending money orders

Time allocation 3 lessons

Suggested teaching, learning and assessment resources

Sample money orders, chart with commission rates, telegram fee, braille, raised pictures/diagrams, mirror, feathers.

Instructions

1. Let learners be in groups.
2. Discuss with learners how to fill in the sample money order.
3. Let learners discuss the following:

   Find how much it would cost to send K1,200.00 by money order as registered item.

4 Let learners present their work.

5 Discuss with learners the working sequence as follows:

Amount	=	K1,200.00
Commission	=	K150.00
Registration	=	K25.00
Total cost	=	Amount + Commission + Registration
	=	K1,200.00 + K150 + K25
	=	K1,375.00

6 Discuss the example in the learners' book.

7 Let learners do exercise 14D from the learners book.

Activity 6 Calculating charges for sending money by telegram

Time allocation 2 lessons

Suggested teaching, learning and assessment resources

Sample money orders, braille, raised pictures, mirror, feathers.

Instructions

1 Let learners be in groups.

2 Let learners discuss the following: John sent K3,300.00 by telegram. Find the total charge.

3 Let them report their work.

4 Discuss with learners the working sequence as follows:

Amount	= K3,300.00
Commission	= K330.00
Telegraph fee	= K20.00
Total cost	= Amount + commission + Telegraph fee
	= K3,300.00 + K330.00 + K20.00
	= K3450.00

5 Let learners do exercise 14E from the learners book

Activity 7 Calculating charges for sending message by telegram

Time allocation 2 lessons

Suggested teaching, learning and assessment resources

Sample telegram forms, braille, raised pictures, mirror, feathers.

Instructions

1. Let learners be in groups.
2. Let learners discuss the following example:

   Calculate how much it would cost to send the following message:

TO:	PETER PHIRI BOX 67 MZUZU
	I NEED TRANSPORT MONEY SCHOOL TERM ENDS ON 17 NOVEMBER
FROM:	JULIET PHIRI PHWEZI GIRLS SECONDARY SCHOOL BOX 2 RUMPHI

3. Let learners present their work.
4. Discuss with learners the working sequence as follows:

   Count number of words with learners then work out charges as follows:

   Number of words = 24

   Rate per word by teleprinter = K3.00.

   Total cost = 24 x 3

   = K72.00

5. Discuss the example in the learners' book.
6. Let learners do exercise 14F from the learners' book

Assessment

Use the scale below to assess each learner.

Name of the learner: ______________________________

Is the learner able to:	Yes	No
1 calculate charges for sending letters?		
2 calculate charges for sending registered items?		
3 calculate commission and total charges for buying and sending postal orders?		
4 calculate commission and charges for buying and sending money orders?		
5 calculate charges for sending money by telegram?		
6 calculate charges for sending message by telegram?		

UNIT 15 Bank services

Time allocation 4 lessons

Introduction

In unit 14 learners were introduced to some of the services that are rendered by the post office. In this unit leaners will be introduced to bank services. Banks are very essential in everyday life because they offer services such as keeping people's money safe and lending money to individuals and organisations.

Success criteria

Learners must be able to describe bank services

Developmental areas

Skills

Ensure that learners acquire the following skills:

- filling in deposit slips
- filling in withdrawal forms

Concepts and knowledge

Ensure that learners understand and acquire the following concepts and knowledge:

- bank services
- deposit slip
- withdrawal form

Attitudes and values

Ensure that learners appreciate the different services offered by banks.

Special needs education

Ensure that the activities are adapted for learners with special educational needs. This will help them develop their potentials and become independent citizens. Observe and record their performance.

Background information

Banks are institutions that among other services, keep money, lend money to organisations and individuals and facilitate bill payments for organisations.

In order for the bank to keep money for someone, one has to open an account with the bank.

When one wants to deposit money, a deposit form is used. There are however, two types of deposit forms, cheque and cash deposit forms.

One will need to fill withdrawal forms in the process of accessing one's money. In some cases, Auto Teller Machine (ATM) cards and cheques are used.

It is therefore important that learners are assisted on how to fill the forms accurately and identify some bank services.

Activity 1 Listing services rendered by the bank

Time allocation 1 lesson

Suggested teaching, learning and assessment resources

Deposit forms, withdrawal forms, sample cheques, samples of Auto teller cards, sample deposit forms, sample withdrawal forms, braille, raised pictures/diagrams

Instructions

1 Let learners be in groups.
2 Ask learners to list down banks that operate in Malawi
3 Ask learners to list down services offered by banks.
4 Let learners present their work.
5 Discuss with learners to establish that banks offer the following services:
   - keeping money safe
   - lending money
   - facilitating payment of bills

Activity 2 Filling in deposit and withdrawal forms

Time allocation 3 lessons

Suggested teaching, learning and assessment resources

Samples of deposit and withdrawal forms, learners' experiences, braille, raised pictures/diagrams

Instructions

1 Let learners be in groups.
2 Provide each group with sample deposit forms.
3 Ask learners to study the deposit forms and discuss how to fill the forms.
4 Let each group present their work.
5 Discuss with learners how to fill a deposit form.
6 Repeat instructions 1 to 5 using a withdrawal form
7 Let learners practise filling in different types of deposit and withdrawal forms.
8 Let learners do exercise 15A from the learners' book.

Assessment

Use the scale below to assess each learner.

Name of the learner: ___________________________

Is the learner able to:	**Yes**	**No**
1 list services rendered by the bank?		
2 fill in deposit forms?		
3 fill in withdrawal forms?		

UNIT 10 Simple interest

Time allocation 12 lessons

Introduction

Interest is sometimes involved whenever money is borrowed or lent out. Learners may be familiar with loans that business people get and the interest charged on the money borrowed. There are two major types of interest. These are compound and simple interest. In this unit, learners will be introduced to simple interest.

Success criteria

Learners must be able to work out simple interest.

Developmental areas

Skills

Ensure that learners develop the following skills:

- calculating simple interest given rate, principal and time
- calculating amount given principal and interest
- calculating principal, given interest, rate and time
- calculating time given interest, principal and rate
- calculating rate given principal, time and interest

Knowledge and concepts

Ensure that learners acquire the following concepts and knowledge:

- simple interst
- principal
- rate
- time
- amount

Attitudes and values

Ensure that learners appreciate the importance of simple interest.

Special needs education

Ensure that the activities are adapted for learners with special educational needs. This will help them develop their potentials and become independent citizens. Observe and record their performance.

Background information

In some cases people agree to pay an extra amount of money in addition to the amount borrowed over a period of time. This extra amount is called interest. For example, if you deposit money in the bank, the bank will pay you interest for the use of that money. If you borrow money, the bank will charge you interest for the use of that money. The sum of money you deposit in the bank or borrow from the bank is called Principal. While the sum of the Principal and Interest is called the Amount. In calculating the simple interest, the following formula is used:

Simple Interest (I) $= \dfrac{PTR}{100}$, where I = interest, P = Principal or the money invested or borrowed, T = Time in years and R = Rate per year. The following set of formula may also be useful in working out problems on simple interest.

Amount (A) = P + I, where P = Principal, I = Interest.

Rate (R) = $\dfrac{100 \times I}{P \times T}$, where I = Interest, P = Principal and T = Time.

Time (T) = $\dfrac{100 \times I}{P \times R}$, where I = interest, P = Principal and R = Rate.

Activity 1 Calculating simple interest

Time allocation 2 lessons

Suggested teaching, learning and assessment resources

Chart paper, learners' experiences, braille.

Instructions

- Discuss with learners the following terms: simple interest, principal, rate, time
- Discuss with learners the following example

Calculate the simple interest on K3500 for 3 years at 4% per annum.

Simple Interest (I) $= \dfrac{PTR}{100}$

Principal = K3,500

Time = 3 years

Rate = 4%

$\therefore$ Simple Interest $= \dfrac{K3{,}500 \times 3 \times 4}{100}$

$= K35 \times 3 \times 4$

$= K420.$

- Discuss the example in the learners' book.
- Let learners do exercise 16A from the learners' book.

Activity 2 Calculating amount

Time allocation 2 lessons

Suggested teaching, learning and assessment resources

Learners' experiences, checklist, braille.

Instructions

1 Let learners discuss the meaning of amount

2 Discuss with learners how to find the amount in the following example:
Find the total amount to be paid back on a loan of K5, 000 for 6 years at 2% simple interest.

$$\text{Amount} = \text{Principal} + \text{Interest}$$

$$\text{Principal} = \text{K}5{,}000.00$$

$$\text{Simple Interest} = \frac{PTR}{100}$$

$$= \frac{\text{K}5,000 \times 6 \times 2}{100}$$

$$= \text{K}600$$

$$\therefore \text{Amount} = \text{K}5,000 + \text{K}600$$

$$= \text{K}5,600$$

3 Discuss the example in the learners' book.

4 Let learners do exercise 16B from the learners' book

Activity 3 Calculating rate

Time allocation 2 lessons

Suggested teaching, learning and assessment resources

Learners' experiences, checklist, braille.

Instructions

1 Discuss with learners how to calculate rate as follows:
At what rate will K2, 500 yield an interest of K250 in 2 years?

$$\text{Rate (R)} = \frac{100 \times I}{P \times T}$$ where I = Interest, P = Principal and T = Time.

Interest = K250, Principal = K2 500, Time = 2 years

$$\therefore \text{Rate} = \frac{100 \times \text{K}250}{\text{K}2500 \times 2}$$

$$= \frac{250}{50}$$

$$= 5\%$$

2 Let learners in their groups, discuss the example in their books.

3 Let learners do exercise 16C from the learners' book.

Activity 4 Calculating time

Time allocation 2 lessons

Suggested teaching, learning and assessment resources

Learners' experiences, checklist, braille.

Instructions

1 Discuss with learners the following example:

How long does it take to obtain K150 interest on K3, 000 at 2% per annum?

$$\text{Time (T)} = \frac{100 \times I}{P \times R}$$

Interest = K150, Principal = K3, 000, Rate = 2%

$$\therefore \text{Time (T)} = \frac{100 \times K150}{K3,000 \times 2}$$

$$= \frac{15}{6}$$

$$= 2^1/_2 \text{ years.}$$

2 Discuss the example in the learners' book.

3 Let learners do exercise 16D from the learners book.

Activity 5 Calculating principal

Time allocation 2 lessons

Suggested teaching, learning and assessment resources

Learners' experiences, braille.

Instructions

1 Discuss with learners the following example:

Find the principal if K300 interest is obtained in 3 years at 5% per annum.

$$\text{Principal} = \frac{100 \times \text{Interest}}{\text{Rate} \times \text{Time}}$$

Interest = K300, Time = 3 years, Rate = 5%

$$\therefore \text{Principal} = \frac{100 \times K300}{5 \times 3}$$

$$= K20 \times 100$$

$$= K2,000.$$

2 Let learners, in their groups, discuss the example in their books.

3 Let learners do exercise 16E from the learners book.

Review exercise

Time allocation 1 lesson

Let learners do the review exercise at the end of Unit 16.

Assessment

Use the table below to assess each learner

Name of learner: ____

How best is the learner able to:	Excellent	Good	Average	Needs support
1 calculate simple interest given rate, principal, and time				
2 calculate amount given principal and interest				
3 calculate rate given interest, principal and time				
4 calculate time given interest, principal and rate				
5 calculate principal given rate, interest and time				

UNIT 17 Cash and bank accounts

Time allocation 7 lessons

Introduction

People, organisations and clubs keep a record of the money they have made and spent. Keeping such records is very important because it helps to monitor how money collected is used. It also helps to check whether one is making profit or loss in business. Therefore learners need to acquire the knowledge of keeping accounting records.

In this unit learners will be introduced to cash and bank accounts.

Success criteria

Learners must be able to:

- maintain cash account
- maintain bank account

Developmental areas

Skills

Ensure that learners acquire the following skills:

- preparing cash account
- preparing bank account
- balancing accounts

Concepts and knowledge

Ensure that learners understand and acquire the following concepts and knowledge:

- cash account
- bank account
- balance brought down
- balance carried down
- debit, credit

Attitudes and values

Ensure that learners appreciate the importance of keeping accounting records.

Special needs education

Ensure that the activities are adapted for learners with special educational needs. This will help them develop their potentials and become independent citizens. Observe and record their performance.

Background information

A cash account is a record of money received and expenses made. The money received is called income and money spent is called payment. A cash account has two sections: the income and payments sections with dates and amounts indicated in each section.

A bank account is a record of money received and payment made in the bank. The sections and columns are similar to those of the cash account. The account contains details of payments made by cheque and money received and paid into the bank.

Activity 1 Identifying columns of a cash account

Time allocation 1 lesson

Suggested teaching, learning and assessment resources

Charts showing column headings of a cash account, braille, raised pictures/diagrams.

Instructions

1 Let learners be in groups.
2 Let them discuss the importance of keeping a record of money received and spent.
3 Let them report their findings.
4 Discuss with learners to establish the importance of keeping a record of money received and spent and the name given to such a record (expected answer: cash account)
5 Discuss with learners the column headings of a cash account as follows:

Dr **Cr**

Date	Income	Amount	Date	Payment	Amount
		K t			K t

6 Help learners to understand that a cash account has two main sections namely Debit (Dr) section and Credit (Cr) section.
7 Let learners copy the cash account above.

Activity 2 Entering transactions in a cash account

Time allocation 2 lessons

Suggested teaching, learning and assessment resources

Charts, lists of transactions, braille, raised pictures/diagrams.

Instructions

1 Provide learners with some transactions made in any club or association within the school eg Agriculture club:

12/10/07	sold maize at K2,000.00
13/10/07	bought tomato seeds at K200.00
15/10/07	sold groundnuts at K950.00
17/10/07	sold vegetables at K300.00
20/10/07	bought fertilizer at K950.00
22/10/07	bought watering can at K250.00
27/10/07	sold onions at K175.00

2 Discuss with learners how to record these transactions in a cash account as follows:

Dr Cr

Date	Income	Amount		Date	Payment	Amount	
		K	t			K	t
12/10/07	sold maize	2000	00	13/10/07	bought seeds	200	00
15/10/07	sold g/nuts	950	00	20/10/07	bought ferlizer	950	00
17/10/07	sold vegetables	300	00	22/10/07	bought watering can	250	00
27/10/07	sold onions	175	00				

3 Let learners practise recording transactions using the following information: Misesa farmers club made the following transactions during the month of June.

June 2	Sales of maize K9,600.00
June 6	Sales of groundnuts K4,750.00
June 7	Bought watering can K490.00
June 21	Bought seeds K220.00
June 24	Sales of vegetables K490.00
June 26	Bought fertilizer K5,700.00
June 28	Paid water bill K1,000.00

Activity 3 Balancing a cash account

Time allocation 3 lessons

Suggested teaching, learning and assessment resources

Charts, lists of transactions, braille, raised pictures/diagrams.

Instructions

1 Let learners be in groups.
2 Let them study the agriculture club cash account drawn in activity 2.
3 Ask learners how the agriculture club would find cash in hand or amount of cash the club has.
4 Help learners to establish that cash in hand is the difference between total income and total payments.
5 Let learners discuss how to balance a cash account.
6 Let learners present their work giving reasons why they think their account is balancing.
7 Discuss with learners that to balance a cash account, the cash in hand is added to the total payments as shown below:

Dr **Agriculture club cash account** **Cr**

Date	Income	Amount K	Amount t	Date	Payment	Amount K	Amount t
12/10/07	sold maize	2000	00	13/10/07	bought seeds	200	00
15/10/07	sold g/nuts	900	00	20/10/07	bought ferlizer	1900	00
17/10/07	sold vegetables	300	00	22/10/07	bought wateringcan	250	00
27/10/07	sold onions	175	00	31/10/07	Balance c/d	1025	00
		3375	00			3375	00
01/11/07	balance b/d	1025	00				

8 Discuss the example in the learners' book.
9 Let learners do exercise 17A from the learners' book.

Activity 4 Preparing a bank account

Time allocation 3 lessons

Suggested teaching, learning and assessment resources

Charts showing columns of bank account, lists of transactions, braille, raised pictures/diagrams.

Instructions

1. Ask learners to mention the importance of keeping money in the bank.
2. Let learners be in groups.
3. Ask learners to study the columns of a bank account and later compare it with those of cash account.
4. Let them report their findings.
5. Help learners to note that the columns of a bank and cash accounts are the same.
6. Let learners enter the following transactions in bank account and balance it.

1 Sept 2007 opened a bank account with K8,000
3 Sept. 2007 bought second hand dresses at K3,000 and paid by cheque
4 Sept. 2007 sold some dresses at K1,300 and was paid a cheque
17 Sept 2007 withdrew K2,000 for office use
22 Sept 2007 sold dresses at K1,900 cash
27 Sept 2007 paid salaries K2,500 by cheque

7. Let learners present their work.
8. Discuss with learners the working sequence as follows:

Dr **Bank account** **Cr**

Date	Income	Amount		Date	Payment	Amount	
		K	t			K	t
1/9/07	Banked	8000	00	3/9/07	bought dresses	3000	00
				17/9/07	withdrawal	2000	00
				27/9/07	paid salaries	2500	00
				30/9/07	Balance c/d	500	00
		8000	00			8000	00
1/10/07	balance b/d	500	00				

Let learners note that:

- cash paid into the bank is shown as income in the bank account
- all payments of money by cheque are entered on the credit side of the bank account
- all withdrawals of cash are entered on the credit side of the bank account.

9. Help learners to understand that the transaction on 4th September 2007 could not be entered in the bank account at this point because it has not been deposited in the bank.
10. Discuss the example in the learners' book.
11. Let learners do exercise 17B from the learners' book.

Review exercise

Time allocation 1 lesson

Let learners do the review exercise at the end of unit 17.

Assessment

Use the table below to assess each learner

Name of learner: ____________________

How best is the learner able to:	Excellent	Good	Average	Needs support
1 enter transactions in a cash account?				
2 balance a cash account?				
3 enter transactions in a bank account?				
4 balance a bank account?				

UNIT 10

Geometrical shapes

Time allocation 10 lessons

Introduction

The idea of geometrical shapes is not new to standard 7 learners. They have been exposed to a number of geometrical shapes from standard one to six.

In this unit, learners will identify types and properties of triangles and quadrilaterals. They will also identify parts of a circle.

Success criteria

Learners must be able to:

- identify types and properties of triangles
- identify types and properties of quadrilaterals
- describe a circle

Developmental areas

Skills

Ensure that learners acquire the following skills:

- identifying types and properties of triangles and quadrilaterals
- identifying parts of a circle
- drawing circles of different sizes

Concepts and knowledge

Ensure that learners understand and acquire the following concepts and knowledge:

- isosceles, equilateral and scalene triangles
- rectangle, square, rhombus, kite, parallelogram and trapezium
- circle

Attitudes and values

Ensure that learners appreciate the importance and application of properties of quadrilaterals, triangles and circle.

Special needs education

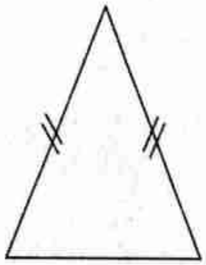
Ensure that the activities are adapted for learners with special educational needs. This will help them develop their potentials and become independent citizens. Observe and record their performance.

Background information

The unit extends the knowledge learners have on geometrical shapes. Three categories of shapes are looked at, these are quadrilaterals, triangles and circle. Learners have been exposed to many objects in the environment with the above mentioned shapes. It would therefore be important to use the local environment when going through the unit.

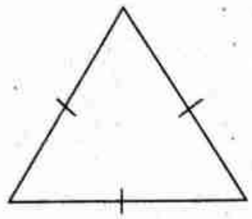
Triangles

A triangle may be defined as a plane figure with three sides and three angles. There are three types of triangles. These are equilateral, isosceles and scalene:



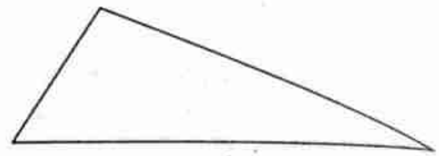
Isosceles triangle

2 sides are equal



Equilateral triangle

All sides are equal

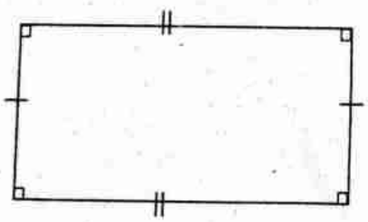


Scalene triangle

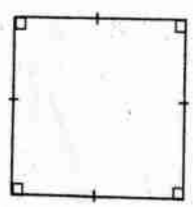
No sides are equal.

Quadrilaterals

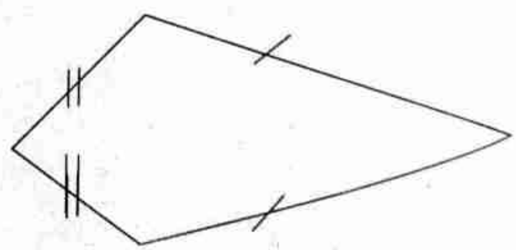
A quadrilateral is a plane figure with four sides and four vertices. There are many types of quadrilaterals. These include; a rectangle, square, rhombus, kite, parallelogram and trapezium.



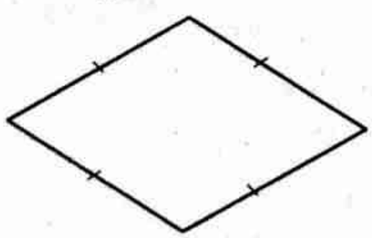
Rectangle



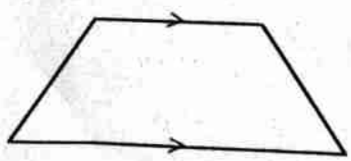
Square



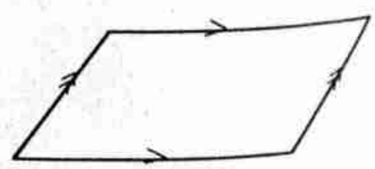
Kite



Rhombus



Trapezium



Parallelogram

Circle

A circle is a plane figure enclosed by a curved line.

Activity 1 Identifying types and properties of triangles

Time allocation 3 lessons

Suggested teaching, learning and assessment resources

Scissors, paper, pencils, observation checklist, rulers, sticks, braille, raised pictures.

Instructions

1. Ask learners to be in groups.
2. Distribute a mixture of shapes of triangles to each group.
3. Let learners sort out the triangles according to shape.
4. Let learners report their findings.
5. Help learners to classify the triangles as follows:
   a with three sides equal
   b with two sides equal
   c with no equal sides
6. Discuss with learners to establish that there are three types of triangles with properties as follows:

Type of triangle	Properties
Equilaterals	• All sides equal • All angles equal
Isosceles	• Two sides equal • Two angles equal
Scalene	• All sides different • All angles different

Activity 2 Identifying types and properties of quadrilateral

Time allocation 4 lessons

Suggested teaching, learning and assessment resources

Different shapes of quadrilaterals, local environment, braille, raised pictures.

Instructions

1 Discuss with learners the meaning of a quadrilateral
2 Let learners be in groups.
3 Ask learners to identify quadrilaterals in the environment.
4 Distribute to each group different shapes of quadrilaterals ie squares, rectangles, parallelograms, rhombus, kites and trapeziums.
5 Let learners sort out quadrilaterals according to shape.
6 Let learners draw the different types of quadrilaterals.
7 Let learners name the types of the quadrilaterals.
8 Discuss with learners to establish the properties of the quadrilaterals as follows:

Type of quadrilateral	Properties
Square	• All sides equal • All angles are right angles
Rectangle	• Opposite sides are equal • All angles are right angles
Parallelogram	• Opposite sides are equal • Opposite angles are equal
Rhombus	• All sides are equal • Opposite angles are equal
Trapezium	• One pair of opposite sides are parallel
Kite	• Adjacent sides are equal • One pair of opposite angles are equal

Activity 3 Identifying parts of a circle

Time allocation 2 lessons

Suggested teaching, learning and assessment resources

Charts, rulers, paper, scissors, pencils, circular objects, pair of compasses, braille, raised pictures.

Instructions

1 Discuss with learners the meaning of a circle
2 Let the learners identify circular objects from the environment
3 Provide each group with circular objects or pairs of compasses.
4 Let learners draw a circle on a sheet of paper.

- Let them fold their circle twice
- Discuss with learners to establish the following in their circle
  a. centre
  b. radius
  c. diameter
  d. circumference
- Assist learners to draw circles of different sizes
- Let learners do exercise 18 in the learners' book

Review exercise

Time allocation 1 lesson

Let learners do the review exercise at the end of unit 18.

Assessment

Use the checklist below to assess each learner

Name of learner: ___________________________

Is the learner able to:	Yes	No
1 identify types of triangles?		
2 identify properties of triangles?		
3 identify types of quadrilaterals?		
4 identify properties of quadrilaterals?		
5 identify parts of a circle?		

Construction of angles

Time allocation 10 lessons

Introduction

The skill of construction is important to learners. It lays a foundation for further education in construction of different shapes. In Standard 6 learners were introduced to angles. They expressed angles using turns, modeled the angles using sticks and papers and identified different types of angles.

In this unit, learners will measure and draw angles using a ruler and a protractor.

Success criteria

Learners must be able to:

- draw angles using a ruler and protractor
- measure angles using a protractor

Developmental areas

Skills

Ensure that learners acquire the following skills:

- measuring angles using a protractor
- drawing angles using a ruler and a protractor

Concepts and knowledge

Ensure that learners understand and acquire the following concepts and knowledge:

- angles
- protractor as an instrument for measuring angles
- degree as a standard unit for measuring angles

Attitudes and values

Ensure that learners appreciate the importance of construction in everyday life.

Special needs education

Ensure that the activities are adapted for learners with special educational needs. This will help them develop their potentials and become independent citizens. Observe and record their performance.

Background information

In Standard 6 learners measured angles using turns. In this class, they will use protractors to draw and measure angles. They will also construct angles using a ruler and a protractor. Construction means drawing accurately. Therefore, it is expected that the learners will be able to draw angles accurately using a sharp pencil, a ruler and a protractor. Since most learners will be using a protractor for the first time, they should be assisted to use it correctly.

Activity 1 Measuring angles

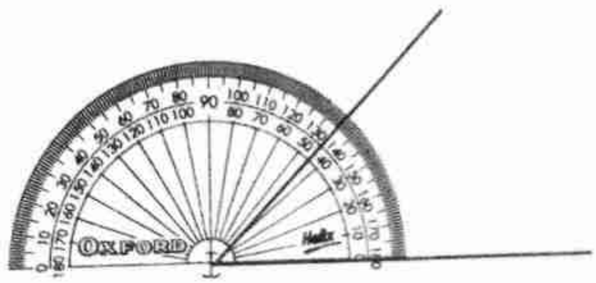
Time allocation 2 lessons

Suggested teaching, learning and assessment resources

Protractors, sharp pencils and rulers, checklist, braille, raised pictures/diagrams.

Instructions

1 Draw two angles of almost the same size.
2 Let learners study the two angles and estimate their sizes.
3 Let them report their estimates.
4 Help learners to establish the need for an instrument for measuring angles.
5 Introduce the protractor to learners as an instrument for measuring angles and a degree (°) as unit for angles.
6 Demonstrate how to measure an angle using a protractor as follows:
First, place the protractor on top of the angle such that the centre of the protractor is on top of the vertex of the angle while the 0° line lies on top of one of the arms of the angle.



The other arm of the angle will then point to the number of degrees marked round the edge of the protractor.
Take the reading as the size of the angle in degrees ie the angle above is 45°.
7 Let learners practice measuring angles using a protractor
8 Let learners do exercise 19A from learner's book.

Activity 2 Drawing angles

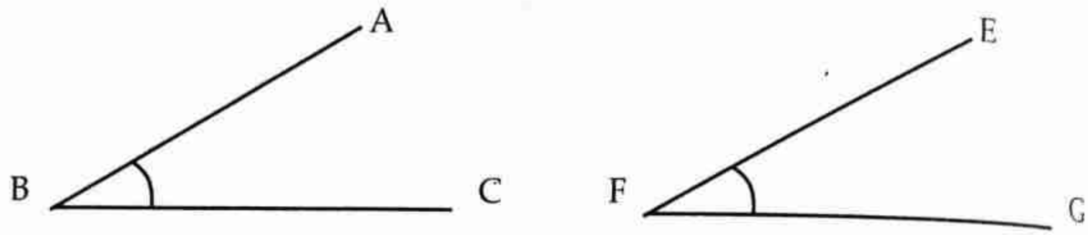
Time allocation 2 lessons

Suggested teaching, learning and assessment resources

Protractors, rulers and pencils, checklist, braille, raised pictures/diagrams.

Instructions

1 Ask learners to draw an angle of any size individually
2 Help learners to establish that to draw an angle, two straight lines should be drawn to meet at a point for example angles marked below are called angle ABC and angle EFG.



3 Help learners to establish that the short form for writing an angle is $\angle$ABC and angle $\angle$ EFG or $A\hat{B}C$ and $E\hat{F}G$.
4 Let learners practice drawing angles
5 Help learners to practice measuring the angles they have drawn

Activity 3 Constructing angles

Time allocation 3 lessons

Suggested teaching, learning and assessment resources

Ruler, protractor and sharp pencils, checklist, braille, raised pictures/diagrams.

Instructions

1 Let learners be in groups.
2 Ask them to construct a 60 angle using a ruler and a protractor.
3 Discuss with learners how to construct a 60 degree angle using a ruler and a protractor.
4 Let learners practice constructing angles using a ruler and a protractor individually.
5 Ask learners to do exercise 19B in the learners' book.

Review exercise

Time allocation 1 lesson

Ask learners to do the review exercise at the end of unit 19.

Assessment

Use the table below to assess each learner

Name of learner: ______________________________

How best is the learner able to:	Excellent	Good	Average	Needs support
1 draw angles using a ruler and a protractor?				
2 measure angles using a protractor?				

UNIT 20 Length

Time allocation 7 lessons

Introduction

In Standard 6 learners were introduced to perimeter of a rectangle and triangle. The idea of perimeter is useful in everyday life. Often people measure distances round objects of different shapes eg finding distance around a rectanglular garden. It is important that learners acquire more knowledge and skills in finding perimeter.

In this unit learners will learn how to find the circumference of a circle and perimeter of composite figures.

Success criteria

Learners must be able to use basic operations to solve practical problems involving length.

Developmental areas

Skills

Ensure that learners acquire the following skills:

- calculating perimeter of circles
- finding perimeter of composite figures
- solving practical problems involving length

Concepts and knowledge

Ensure that learners understand and acquire the following concepts and knowledge:

- circumference of a circle
- perimeter of composite figures

Attitudes and values

Ensure that learners appreciate the importance of perimeter in everyday life.

Special needs education

Ensure that the activities are adapted for learners with special educational needs. This will help them develop their potentials and become independent citizens. Observe and record their performance.

Background information

The circumference of a circle (perimeter) is the length round the circle. To calculate the circumference of a circle learners need to understand the concept of pi (π). Ensure that learners understand the concept of pi through practical activities.

The value of π is approximately equal to 3.14 or $^{22}/_{7}$. The circumference of a circle is therefore calculated by multiplying π by its diameter, or simply πD or $2\pi r$.

On the other hand, the perimeter of composite figures is calculated by adding the lengths of all the outer edges that make up the figure.

Activity 1 Introducing pi

Time allocation 2 lessons

Suggested teaching, learning and assessment resources

Circular and cylindrical objects such as tin of milk, plate, tyre, coin, thread, paper, braille, raised pictures/diagrams.

Instructions

1 Let learners be in groups.
2 Provide each group with circular or cylindrical objects.
3 Ask learners to measure the circumference and diameter of each circular objet using a piece of paper or thread and rulers.
4 Let them divide the circumference by the diameter of each object and record the results in a table as shown below:

eg

Object	**Circumference**	**Diameter**	$\frac{\textbf{Circumference}}{\textbf{diameter}}$
Tin of milk			
Plate			
Tyre			
Coin			

5 Ask learners to study the results especially circumference divided by diameter.
6 Help learners to establish that the circumference of the circle is about 3 times the diameter. This ratio is called pi (π).
7 Help learners to establish that π is approximately 3.14 or $^{22}/_{7}$.

8 Using step 6 and 7 help learners to establish that the formula for circumference of a circle $= \pi \times$ diameter

$= \pi d$

$= 2\pi r$

Activity 2 Calculating circumference of a circle

Time allocation 2 lessons

Suggested teaching, learning and assessment resources

Learners' experiences, checklist, braille, raised pictures/diagrams.

Instructions

1 Let learners be in groups.

2 Ask them to work out the following:

Find the circumference of a circle whose diameter is 21cm.

3 Let learners report their work.

4 Discuss with learners the working sequence as follows:

$$\begin{aligned} \text{Circumference of a circle} &= \pi d \\ &= \frac{22}{7} \times 21\text{cm} \\ &= 22 \times 3\text{cm} \\ &= 66\text{cm} \end{aligned}$$

5 Discuss the example in the learners' book

6 Let the learners do exercise 20A from the learners' book.

Activity 3 Calculating perimeter of composite figures

Time allocation 2 lessons

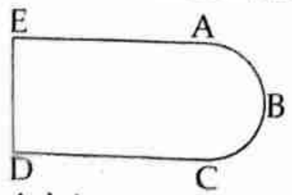
Suggested teaching, learning and assessment resources

Charts showing composite figures, threads, pairs of scissors, ruler, checklist, braille, raised pictures/diagrams.

Instructions

1 Let learners be in groups.

2 Provide larners with charts showing composite figures eg:



3 Ask learners to measure the perimeter of the figure above.

4 Discuss with learners how to calculate the perimeter of the figure as follows:

- let learners identify the sides that make up the distance around the figure as ABC, CD, DE and EA.

$$\begin{aligned} \therefore \text{Perimeter} &= ABC + CD + DE + EA \\ &= {}^{1}/_{2}\pi d + CD + DE + EA \\ &= \frac{1}{2} \times \frac{22}{7} \times 7\text{cm} + 3\text{cm} + 7\text{cm} + 3\text{cm} \\ &= 11\text{cm} + 3\text{cm} + 7\text{cm} + 3\text{cm} \\ &= 24\text{cm} \end{aligned}$$

Note: In finding the distance ABC, we divide by 2 because it is a semicircle (half circle).

5 Let learners do exercise 20B from the learners' book.

Review exercise

Time allocation 1 lesson

Let learners to do the review exercise on unit 20.

Assessment

Use the scale below to assess each learner

Name of learner: ______________________________

How best is the learner able to:	Excellent	Good	Average	Needs support
1 calculate circumference of a circle?				
2 work out perimeter of composite figures?				
3 solve practical problems on perimeter?				

UNIT 21 Mass

Time allocation 6 lessons

Introduction

In Standard six learners were introduced to basic operations involving mass. It is important that learners are exposed to more practical problems on mass. They should be given examples depicting real life situations.

In this unit, learners will work out practical problems involving mass.

Success criteria

Learners must be able to use basic operations to solve practical problems involving mass.

Developmental areas

Skills

Ensure that learners acquire the skills of solving practical problems involving mass

Concepts and knowledge

Ensure that learners develop an understanding of the following concepts and knowledge:

- mass
- addition and subtraction of mass
- multiplication and division of mass
- relationship between kilogram and grams

Attitudes and values

Ensure that learners appreciate the importance of mass.

Special needs education

Ensure that the activities are adapted for learners with special educational needs. This will help them develop their potentials and become independent citizens. Observe and record their performance.

Background information

Learners have already been introduced to standard units for mass. Ensure that learners understand the relationship between grams and kilograms ie 1,000gm = 1kg.

Activity 1 Carrying out basic operations involving mass

Time allocation 3 lessons

Suggested teaching, learning and assessment resources

Cards, checklist, braille, raised pictures/diagrams.

Instructions

1 Let learners be in groups.
2 Ask learners to work out the following:

i

	kg	gm
	625	247
	122	335
+	204	726

ii

	kg	
	35 .	72
-	28 .	36

3 Let them report their work.
4 Discuss with learners the working sequnce as follows:

i

	kg		gm
	625		247
	122		335
+	204		726
	952		308
	1	←	1
	$\underline{951}$		
	952	$\underline{1000}$	1308
		-	$\underline{1000}$
			308

ii

	kg	
	35 .	72
-	28 .	36
	7 .	36

5 Discuss the example in the learners' book.
6 Let learners do Exercise 21A from the learners' book.

Activity 2 Solving practical problems involving mass

Time allocation 2 lessons

Suggested teaching, learning and assessment resources

Cards, checklist, braille, raised pictures/diagrams.

Instructions

1 Let learners be in groups.
2 Ask learners to work out the following problem.
An organisation distributed 3555kg of flour to 15 families. How much flour did each family get?
3 Let them report their work.
4 Discuss with learners the working sequnce as follows:
3,555kg ÷ 15

$$
\begin{array}{r|l}
 & \text{kg} \\
 & 237 \\ \cline{2-2}
15 & 3,555 \\
 & \underline{30} \\
 & 55 \\
 & \underline{45} \\
 & 105 \\
 & \underline{105} \\
 & 0
\end{array}
$$

= 237kg
= 237kg

5 Let learners do Exercise 21B from the learners' book.

Review exercise

Time allocation 1 lesson

Let learners turn to the learners' book unit 21 and do the the review exercise.

Assessment

Use the scale below to assess each learner

Name of learner: ______________________

How best is the learner able to:	Excellent	Good	Average	Needs support
1. add mass?				
2. subtract mass?				
3. multiply mass?				
4. divide mass?				

UNIT 22 Time intervals

Time allocation 7 lessons

Introduction

Time intervals are commonly used in real life, for example, in competitions, timetables, regulating events. In mathematics and other subjects, certain terms are used to describe a period or duration of time eg: weekly, annually, etc. It is therefore, important that learners must be introduced to these terms for easy communication.

In this unit, learners will describe time intervals and develop time lines.

Success criteria

Learners must be able to:

- describe time intervals
- develop time lines

Developmental areas

Skills

Ensure that learners acquire the following skills:

- describing time intervals
- developing time lines
- measuring time intervals

Concepts and knowledge

Ensure that learners develop an understanding of the following concepts and knowledge

- describing time intervals
- interpreting time lines
- solving practical problems on time intervals

Attitudes and values

Ensure that learners appreciate the importance of time intervals.

Special needs education

Ensure that the activities are adapted for learners with special educational needs. This will help them develop their potentials and become independent citizens. Observe and record their performance.

Background information

Short time intervals have already been introduced to learners in the lower classes. For example, the second, the minute, the hour, a day, week, month and year.

In this unit, learners will be introduced to other types of time intervals. They will discuss different terms used to describe events and their meanings. Some of the terms and their meanings are as follows:

fortnight	-	14 days (2 weeks)
olympiad	-	4 years
instrum	-	5 years
decade	-	10 years
score	-	20 years
silver jubilee	-	celebrations after 25 years
golden jubilee	-	celebrations after 50 years
human generation	-	30 years
diamond jubilee	-	celebrations after 60 years
century	-	100 years
centenary	-	celebrations after 100 years
millennium	-	1,000 years

Learners will also be introduced to time lines, ie the time intervals in relation to the time before and after the birth of Jesus Christ. The time interval before the birth of Jesus Christ is denoted BC while AD refers to the time interval after the birth of Jesus Christ.

AD stands for Anno Dominni (latin word) and BC stands for Before Christ's birth. Usually a number line is used to explain AD and BC concepts for easy understanding of the same by learners.

It is hoped that the learners will be actively involved so that this unit does not appear informative.

Activity 1 Describing time intervals up to 50 years

Time allocation 3 lessons

Suggested teaching, learning and assessment resources

Charts of time intervals, checklist, braille, raised pictures/diagrams, mirror, feathers.

Instructions

1 Let learners be in groups.

2 Distribute charts with the following time intervals to learners:

Time intervals	Meaning
fortnight	
bimonthly	
annually	
decade	
silver jubilee	
golden jubilee	
human generation	

3 Let learners discuss meanings of the terms in the table above.

4 Let learners present their work.

5 Help learners to establish the meanings as follows:

Time intervals	Meaning
fortnight	14 days (2 weeks)
bimonthly	every 2 months
annually	yearly
decade	10 years
silver jubilee	celebrations after 25 years
golden jubilee	celebrations after 50 years
human generation	30 years

6 Write the following on the chalkboard:

Mrs Phiri will hold her silver-jubilee celebrations next year. How old is she now?

7 Discuss with learners how Mrs Phiri's age can be calculated as follows:

silver jubilee = 25 years

time from now to next year = 1 year

therefore Mrs Phiri's age = (25 - 1) years

= 24 years

Therefore Mrs Phiri is 24 years old

8 Discuss the examples in the learners' book.

9 Let the learners do Exercise 22A from the learners' book.

Activity 2 Describing time intervals from 60 years up to 1,000 years

Time allocation 2 lessons

Suggested teaching, learning and assessment resources

Charts of time intervals, checklist, braille, raised pictures/diagrams.

Instructions

1 Let learners be in groups.

2 Display a chart of time intervals as shown below:

Time intervals	Meaning
diamond-jubilee	
a century	
centinary	
millenium	

3 Let learners discuss meanings of these terms in relation to celebrations that are held in their communities.

4 Let learners present their work.

5 Help learners to establish the meaning of the terms above.

6 Discuss with learners the following example:

How many centuries are there in a millenium?

Millenium = 1,000 years

Century = 100 years

$\therefore$ Number of centuries in millenium $= \dfrac{1,000 \text{ years}}{100 \text{ years}}$

$= 10$ centuries

7 Discuss the example in the learners' book.

8 Let learners do exercise 22B from the learners' book.

Activity 3 Developing time lines

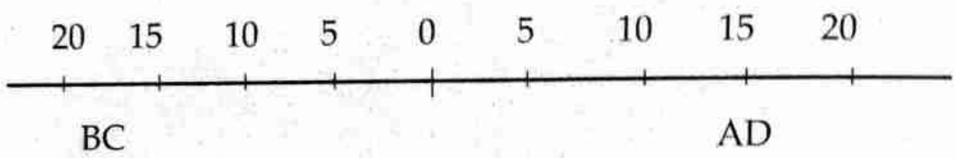
Time allocation 1 lesson

Suggested teaching, learning and assessment resources

Time line chart, checklist, braille, raised pictures/diagrams.

Instructions

1 Let learners be in groups.
2 Display the chart of time line as shown below:



3 Let learners discuss the meanings of the terms BC and AD.
4 Let learners present their work.
5 Help learners to establish that BC means Before Christ and AD means After the Birth of Christ (Anno Dominni).
6 Let learners locate the following on the line: 5BC, 5AD, 15AD.
7 Discuss with learners the number of years between 10BC and 15AD on the time line as follows:

Number of years = 10 years + 15 years
= 25 years

8 Discuss the example in the learners' book
9 Let the learners do Exercise 22C from the learners' book.

Review exercise

Time allocation 1 lesson

Let the learners turn to the learners' book unit 22 and do the review exercise.

Assessment

Use the scale below to assess each learner

Name of learner: ____________________

How best is the learner able to:	Excellent	Good	Average	Needs support
1 describe time intervals?				
2 interpret time intervals?				
3 measure time intervals?				

UNIT 23 Area

Time allocation 8 lessons

Introduction

In Standard 6 learners calculated areas of rectangles, squares and triangles. Area is a useful concept in everyday life. It is applied in several fields such as agriculture and construction industry. Therefore learners need to learn and acquire more knowledge and skills in finding area.

In this unit, learners will be introduced to area of parallelograms, circles and composite figures.

Success criteria

Learners must be able to find area of parallelograms and circles.

Developmental areas

Skills

Ensure that learners acquire the following skills:

- establishing formula for finding area of a parallelogram and a circle
- finding area of a parallelogram
- finding area of a circle

Concepts and knowledge

Ensure that learners develop an understanding of the following concepts and knowledge:

- area of a parallelogram = base x height
- area of a circle = πr^2
- area of a composite figure = sum of areas of the shapes that make up the figure

Attitudes and values

Ensure that learners appreciate the importance of area.

Special needs education

Ensure that the activities are adapted for learners with special educational needs. This will help them develop their potentials and become independent citizens. Observe and record their performance.

Background information

By now, learners should know how to calculate the area of a rectangle and a triangle. In this unit they will learn how to find the area of a parallelogram, a circle and a composite figure. Learners will explore how to derive the formula for finding the area of a parallelogram and a circle.

A composite figure is a figure that is formed by combining two or more different shapes or figures. Its area is found by adding the areas of individual shapes that make the figure. For instance if the composite figure is made up of a rectangle and a triangle then its area will be equal to area of the rectangle plus area of the triangle. It is therefore important that learners start by identifying the shapes that make up the composite figure in order to find their areas.

Activity 1 Finding area of a parallelogram

Time allocation 2 lessons

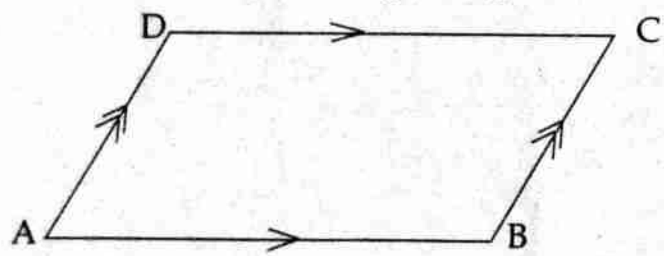
Suggested teaching, learning and assessment resources

Charts, checklist, braille, pencils, rulers.

Instructions

1 Let learners be in groups.

2 Let learners draw a parallelogram eg:



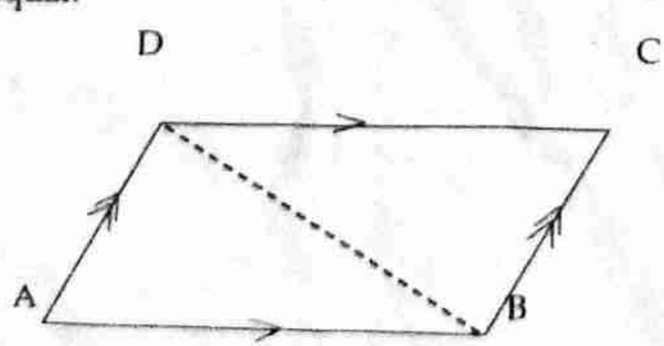
3 Ask learners the formula for finding the area of a triangle.

4 Ask learners to discuss the formula for the area of the parallelogram.

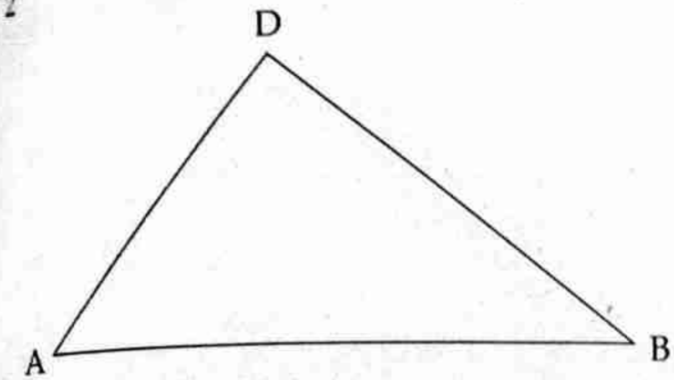
5 Let them report their work.

6 Discuss with learners the formula for the area of a parallelogram as follows:

- Let learners draw the line BD (ie the diagonal). Help learners to understand that by joining BD two triangles have been formed and the two triangles are equal.



- discuss with learners how to find area of triangle ABD. (Expected answer is $\frac{1}{2}$ base x height.)



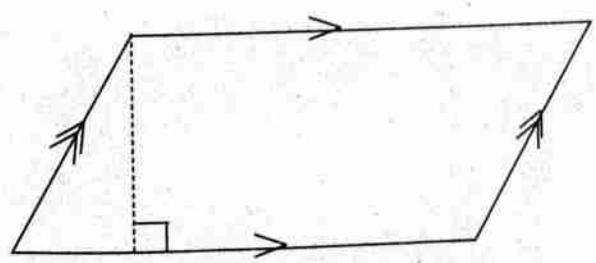
- Help learners to establish that the area of the parallelogram ABCD is twice the area of triangle ABD ie $2 \times \frac{1}{2}$ base x height

= base x height

∴ area of a parallelogram = base x height

Note: height in this case is the perpendicular height (at 90° to the base).

- Discuss with learners how to find the area of the parallelogram given below:



7 Discuss with the learners how to find the area of the parallelogram as follows:

Area of parallelogram = base x height
= 10 cm x 9 cm
= $90cm^2$

8 Discuss the example in the learners' book.

9 Let the learners do Exercise 23A from the learners' book.

Activity 2 Finding area of a circle

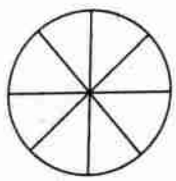
Time allocation 2 lessons

Suggested teaching and learning resources

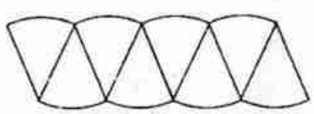
Paper, a pair of scissors, chart paper, checklist, braille, raised diagrams.

Instructions

1. Ask learners to be in groups.
2. Ensure that each group has the following resources: razor blade/scissors, a piece of paper and a pencil.
3. Demonstrate the steps for establishing the formula for the area of a circle as follows:
   - Divide the circle into a number of equal parts as shown below:



   - Cut the circle along the marked lines and arrange the pieces as shown below:



   - Discuss with learners to establish that the figure formed is a parallelogram
   - Discuss with learners the base and the height of this parallelogram.
4. Help the learners to establish that
   i the base of this parallelogram is 1/2 the circumference of the circle

   $= \frac{1}{2}$ of $2\pi r \quad = \pi r$

   ii the height is the width which is r

   iii therefore the area of the parallelogram above

   = base x height

   $= \pi r \times r$

   $= \pi r^2$

$\therefore$ the area of a circle is equal to πr^2.

5 Discuss with learners the following example:
Find the area of the following circle

7cm radius

$$\text{Area of a circle} = \pi r^2$$
$$= \frac{22}{7} \times 7\text{cm} \times 7\text{cm}$$
$$= 154\text{cm}^2$$

6 Let the learners do exercise 23B from the learners' book.

Activity 3 Finding area of composite figures

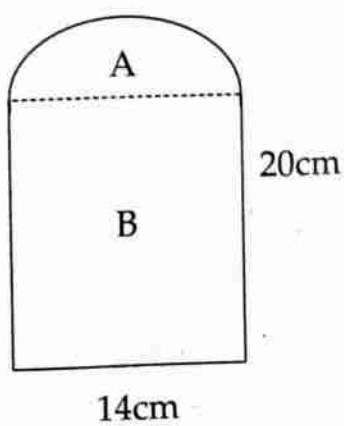
Time allocation 3 lessons

Suggested teaching, learning and assessment resources

Charts showing composite figures, checklist, braille, raised pictures/diagrams.

Instructions

1 Ask learners the formula for finding areas of rectangles, triangles and circles.
2 Ask learners to identify shapes that make up the composite figures shown on the chart eg:



3 Discuss with learners that the figure is made up of a rectangle and a semi-circle (half a circle).
4 Let learners discuss how area of such a figure could be calculated.

5 Help learners to establish that the area of the figure could be found by adding area of the rectangle and area of the semi-circle as follows:

$$\text{Area of rectangle (A)} = L \times B$$
$$= 20cm \times 14cm$$
$$= 280cm^2$$

$$\text{Area of a circle} = \pi r^2$$

$$\therefore \text{ area of the semi-circle (B)} = \frac{\pi r^2}{2}$$
$$= \frac{3^1/_7 \times 7cm \times 7cm}{2}$$
$$= \frac{22}{7} \times \frac{7cm \times 7cm}{2}$$
$$= 77cm^2$$

$$\text{Area of the figure} = \text{area A} + \text{area B}$$
$$= 280cm^2 + 77cm^2$$
$$= 357cm^2$$

6 Let learners do exercise 23C from the learners' book.

Review exercise

Time allocation 1 lesson

Let learners turn to the learners' book and do the review exercise of unit 23.

Assessment

Use the scale below to assess each learner

Name of learner: ______________________________

How best is the learner able to:	Excellent	Good	Average	Needs support
1 establish formula for finding area of a parallelogram and a circle?				
2 establish formula for finding area of a circle?				
3 find area of a parallelogram?				
4 find area of a circle?				
5 find area of a composite figure?				

UNIT 24 Volume and capacity

Time allocation 8 lessons

Introduction

Learners have already been introduced to volume and capacity. In this unit learners will be introduced to the relationship between volume and capacity.

Success criteria

Learners must be able to find the volume of cubes and cuboids.

Developmental areas

Skills

Ensure that learners acquire the following skills:

- converting volume to capacity and vice versa
- finding volume of cuboids and cubes using formula

Concepts and knowledge

Ensure that learners understand and acquire the concept and knowledge of volume and capacity.

Attitudes and values

Ensure that learners appreciate the relationship between volume and capacity.

Special needs education

Ensure that the activities are adapted for learners with special educational needs. This will help them develop their potentials and become independent citizens. Observe and record their performance.

Background information

Volume and capacity are not one concept. Volume is the amount of space occupied by an object while capacity is the ability of a container to hold a substance. Volume is measured in cubic metres, (m^3) while capacity is measured in litres. It is a good idea to give learners adequate practice on volume and capacity so that they appreciate the difference. However, 1 litre is equivalent to $1{,}000cm^3$.

Activity 1 Converting volume to capacity, and capacity to volume

Time allocation 2 lessons

Suggested teaching, learning and assessment resources

Chart showing volume and capacity relationships, sets of cards, braille.

Instructions

1. Let learners be in groups
2. Ask learners to study the relationship between volume and capacity as follows:

Volume	Capacity
$1000cm^3$	1 litre (1ℓ)
$1\ cm^3$	1 millilitre (1mℓ)

3. Ask learners to convert:
   a. $4000cm^3$ to litres.
   b. 240mℓ to cm^3
4. Let learners present their work.

Volume	Capacity
$4000\ cm^3$	? ℓ
? cm^3	240 mℓ

5. Discuss with learners the working sequence as follows:

a. $1{,}000cm^3 = 1$ litre

$$4{,}000cm^3 = \frac{4000cm^3}{1000cm^3} \times 1 \text{ litre}$$

$$= 4 \text{ litres}$$

b. $1ml = 1cm^3$

$$240ml = \frac{240m\ell}{1m\ell} \times 1cm^3$$

$$= 240cm^3$$

6. Discuss the examples in the learners' book.
7. Let learners do exercise 24A from the learners' book.

Activity 2 Finding volume of cuboids and cubes using formula

Time allocation 2 lessons

Suggested teaching, learning and assessment resources

Cubes, cuboids, chart showing formula for calculating volume, rulers, braille.

Instructions

1. Let learners be in groups
2. Ask learners to work out the volume of the following:

3 Discuss with learners the working sequence as follows:

a. $V = L \times W \times H$
$= 10 \times 8 \times 9$
$= 720\text{cm}^3$

b. $V = L \times W \times H$
$= 5 \times 5 \times 5$
$= 125\text{cm}^3$

4 Discuss the examples in the learners' book.

5 Let learners do exercise 24B from the learners' book

Activity 3 Solving practical problems on volume and capacity

Time allocation 2 lessons

Suggested teaching, learning and assessment resources

Learner's book, cubes, cuboids, chart showing formula for calculating volume, rulers, braille.

Instructions

1 Discuss with learners the following examples

Example 1

A bottle contains 300 millilitres of water. If half of it is drank, find the volume of the remaining water in cm^3

$$\text{Amount of water left} = \frac{300\text{m}\ell}{2}$$
$$= 150\text{m}\ell$$
$$1\text{m}\ell = 1\text{cm}^3$$
$$150\text{m}\ell = \frac{150\text{m}\ell}{1\text{m}\ell} \times 1\text{cm}^3$$
$$= 150\text{cm}^3$$

Example 2

A tank which is 8m long, 6m high and 5m wide is full of water. Find the volume of water in the tank.

$$\text{Volume of water in the tank} = 8\text{m} \times 6\text{m} \times 5\text{m}$$
$$= 240\text{m}^3$$

2 Let learners do exercise 24C from the learners' book.

Review exercise

Time allocation 1 lesson

Let learners turn to the learners' book unit 24 and do the review exercise.

Assessment

Use the table below to assess each learner

Name of learner: ______________________________

Is the learner able to:	Yes	No
1 convert volume to capacity?		
2 convert capacity to volume?		
3 find volume of cuboids and cubes using formula?		
4 solve practical problems on volume?		
5 solve practical problems on capacity?		

UNIT 25 Temperature

Time allocation 7 lessons

Introduction

In Standard 6, learners were introduced to the idea of temperature.

In this unit, learners will estimate and measure temperature of substances. Learners will also have the opportunity to analyze information on temperature and make interpretations.

Success criteria

Learners must be able to:

- estimate temperatures of substances
- measure temperatures of substances

Developmental areas

Skills

Ensure that learners develop the following skills:

- estimating and measuring temperature
- collecting and recording data on temperature
- presenting data on temperature into tables and bar graphs
- interpreting data on temperature from tables and bar graphs

Concepts and knowledge

Ensure that learners acquire the following concepts and knowledge:

- estimation of temperature
- presentation of data on temperature into tables and bar graphs
- interpretation of data on temperature from tables and bar graphs

Attitudes and values

Ensure that learners appreciate the importance of:

- estimating and measuring temperature
- presenting data on temperature into tables and bar graphs
- interpreting data on temperature from tables and bar graphs

Special needs education

Ensure that the activities are adapted for learners with special educational needs. This will help them develop their potentials and become independent citizens. Observe and record their performance.

Background information

In this unit, learners will estimate and measure temperature. They will present data on temperature into tables and bar graphs. They will also interpret information on temperature from the tables and bar graphs.

Learners will find this unit motivating if the ideas and examples discussed are drawn from their everyday experiences. Learners often confuse bar graphs with histograms whose bars touch each other to show continuity. You might find it necessary to refer them to standard 6 work as a reminder.

Activity 1 Estimating and measuring temperatures of water

Time allocation 3 lessons

Suggested teaching and learning resources

Thermometers, cold water, warm water, hot water, containers such as tins, bottles, braille.

Instructions

1 Let learners be in groups
2 Distribute thermometers, cold water, warm water and hot water.
3 Ask learners to estimate and then measure the temperature of the water provided.
4 Let learners record their findings in a table as follows:

Types of water	Estimated temperature	Actual temperature
Cold water		
Warm water		
Hot water		

5 Ask learners to identify estimates that are so close to the actual temperature.

Activity 2 Presenting data into tables and bar graphs

Time allocation 3 lessons

Suggested teaching and learning resources

You will need pencils and rulers, braille.

Instructions

1 Let learners be in groups
2 Ask learners to construct a table that will show type of water and actual temperature

Types of water	Actual temperature (°C)
Cold	
Warm	
Hot	

3 Discuss with learners how to present information in the table in form of bar graph.
4 Discuss the example in the learners' book.
5 Let learners do Exercise 25A from the learners' book.

Activity 3 Interpreting data from tables and bar graphs

Time allocation 1 lesson

Suggested teaching and learning resources

Charts, data, pencils and rulers, braille.

Instructions

1 Let learners be in groups
2 Provide the following information to learners:

Daily average temperature readings recorded at 5 weather stations in Malawi are as follows:

Weather station	Temperature reading (°C)
A	26
B	18
C	32
D	41
E	14

3 Ask learners to draw the table in their exercise books and use the information to construct bar graphs
4 Ask learners to identify the
   a. coldest weather station
   b. hottest weather station
5 Discuss the example in the learners' book
6 Let learners do Exercise 25B from the learners' book

Assessment

Use the scale below to assess each learner

Name of learner ____________________

How best is the learner able to:	Excellent	Good	Average	Needs support
1 establish temperature of substances?				
2 mesure temperature of substances?				

Patterns

Time allocation 4 lessons

Introduction

Ideas of number and geometric patterns were introduced in previous classes. In this class learners will be provided with more knowledge and skills that will be helpful in everyday life. Further, the knowledge and skills gained will form a foundation for designing and for learning secondary school mathematics.

In this unit, learners will extend number and geometric patterns.

Success criteria

Learners must be able to:
- investigate and extend number patterns
- investigate and extend geometric patterns

Developmental areas

Skills

Ensure that learners develop the following skills:
- identifying number and geometric patterns
- extending number and geometric patterns
- establishing the rules of number and geometric patterns

Concepts and knowledge

Ensure that learners acquire the following concepts and knowledge:
- number patterns
- geometric patterns

Attitudes and values

Ensure that learners appreciate the importance of number and geometric patterns.

Special needs education

Ensure that the activities are adapted for learners with special educational needs. This will help them develop their potentials and become independent citizens. Observe and record their performance.

Background information

By now, learners should be able to extend number patterns by adding or subtracting a particular number to the previous term to form the next term. In this unit, they will deal with number patterns whose terms are formed by multiplying or dividing the previous term by a particular number. They will also extend geometric patterns.

Activity 1 Number patterns

Time allocation 2 lessons

Suggested teaching, learning and assessment resources

Chart showing number patterns, learners' experiences, braille.

Instructions

1 Let learners be in groups.

2 Ask learners to establish the rule involved in the following number pattern and work out the next three numbers.

3, 6, 12, 24, ___, ___, ___

3 Let learners report their work.

4 Help learners to establish that the rule is multiply the previous number by 2 to get the next number.

5 Discuss the examples in the learners' book.

6 Let the learners do Exercise 26A in the learners' book.

Activity 2 Extending geometric patterns

Time allocation 2 lessons

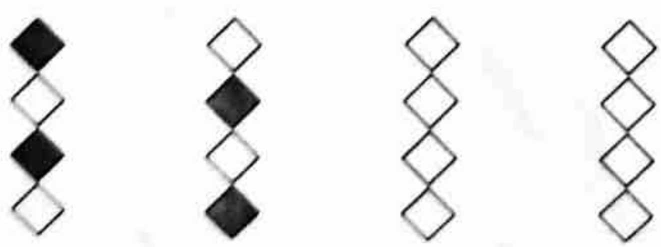
Suggested teaching, learning and assessment resources

Chart showing geometric patterns, checklist, bottle tops, braille.

Instructions

1 Let learners be in groups.

2 Ask learners to draw or make the next two shapes in the following pattern:



3 Let learners display their shapes.

4. Discuss with learners to establish the rule involved in the pattern.
5. Let the learners do exercise 26B from the learners' book.

Review exercise

Time allocation 1 lesson

Let learners do the review exercise at the end of unit 26.

Assessment

Use the scale below to assess each learner

Name of learner : ______________________________

How best is the learner able to:	Yes	No
1 identify number patterns?		
2 identify geometric patterns?		
3 extend number patterns?		
4 extend geometric patterns?		
5 establish the rule for finding patterns?		

UNIT 27 Addition and subtraction of like terms

Time allocation 4 lessons

Introduction

Although learners have been introduced to the algebraic language from Standard 5, they have not practiced the language adequately. This unit will give them an opportunity to practice addition and subtraction of like terms. This knowledge will lay foundation for secondary school mathematics.

In this unit, learners will add and subtract like terms.

Success criteria

Learners must be able to:

- add like terms
- subtract like terms

Developmental areas

Skills

Ensure that learners acquire the following skills:

- adding like terms
- subtracting like terms

Concepts and knowledge

Ensure that learners understand and acquire the following concept and knowledge of like terms.

Attitudes and values

Ensure that learners appreciate the importance of addition and subtraction of like terms.

Special needs education

Ensure that the activities are adapted for learners with special educational needs. This will help them develop their potentials and become independent citizens. Observe and record their performance.

Background information

Like terms are terms that have the same letters or variables for example in the expression $2a - 5a - 3t + 7t$: $2a$ and $5a$ or $3t$ and $7t$ are like terms. The letters can represent any number of objects. Addition and subtraction of like terms is similar to addition and subtraction of whole numbers.

Activity 1 Adding and subtracting like terms

Time allocation 3 lessons

Suggested teaching, learning and assessment resources

Objects such as bananas and apples, learners' experiences, braille, raised pictures/diagrams.

Instructions

1 Show learners the following expression.
  $4x + 5m + 6x + 9m$
2 Ask learners to identify terms that have the same letters.
3 Let them report their work.
4 Assist learners to establish that in the expression $4a + 5b + 6a + 9b$, $4a$ and $6a$ are like terms.
5 Ask learners to simplify the following:
  a. $3a + 4a + 6a$
  b. $5b - 7b + 4b$
6 Let learners present their work.
7 Discuss with learners how to add or subtract like terms as follows:
  a. $3a + 4a + 6a = 13a$
  b. $5b - 7b + 4b$
  $= 5b + 4b - 7b$
  $= 9b - 7b$
  $= 2b$

8 Let learners do exercise 27 in the learners' book

Review exercise

Time allocation 1 lesson

Let learners do the review exercise at the end of unit 27.

Assessment

Use the scale below to assess each learner

Name of learner: ______________________________

How best is the learner able to:	Excellent	Good	Average	Needs support
1 add like terms?				
2 subtract like terms?				

UNIT 28 Inequalities

Time allocation 7 lessons

Introduction

The concept of inequalities is not new to Standard 7 learners. They were introduced to inequalities in standard 6. They were also exposed to number sentences in Standards 5 and 6, however they completed number sentences up to three terms only.

In this unit they will complete number sentences up to four terms, develop number sentences and use inequality symbols appropriately.

Success criteria

Learners must be able to:

- complete number sentences
- formulate number sentences
- use inequality symbols appropriately

Developmental areas

Skills

Ensure that learners develop the following skills:

- completing number sentences
- formulating number sentences from a given context
- using inequality symbols appropriately

Concepts and knowledge

Ensure that learners acquire the following concepts and knowledge:

- number sentences
- inequalities
- symbols, greater than (>), less than (<), not equal (≠)

Attitudes and values

Ensure that learners appreciate the importance and application of number sentences and inequalities in everyday life.

Special needs education

Ensure that the activities are adapted for learners with special educational needs. This will help them develop their potentials and become independent citizens. Observe and record their performance.

Background information

By now learners are conversant with inequality symbols; not equal to (≠), greater than (>) and less than (<). In this unit, learners are also exposed to number sentences up to four terms. Learners need adequate time to practice solving problems on number sentences and appropriate use of the inequality symbols. Real life examples may assist the learners to internalize the concept.

It is important to guide learners that when solving equations; what is done on one side of the equation should also be done on the other side.

Activity 1 Formulating number sentences

Time allocation 1 lesson

Suggested teaching and learning resources

Learners' experiences, checklist, braille.

Instructions

1 Let learners be in groups.
2 Ask learners to formulate number sentence from the following:
Mary thinks of a number y, increases it by 9, the result is 15.
3 Let learners present their work.
4 Discuss with learners that the number sentence is as follows:
$y + 9 = 15$
5 Let learners do exercise 28A in the learner's book.

Activity 2 Solving practical problems on number sentences

Time allocation 2 lessons

Suggested teaching and learning resources

Learners' experiences, checklist, braille.

Instructions

1 Let learners be in groups.
2 Ask learners to solve the following problem:
There are y bulbs in a house. If 7 were burnt and 11 were in good order, find the total number of bulbs in the house.
3 Let learners present their work.
4 Discuss with learners the working as follows:
$y - 7 = 11$ (Add 7 on both sides)
$y - 7 + 7 = 11 + 7$
$y = 18$
There were 18 bulbs in total
5 Let learners do exercise 28B from the learners' book.

Activity 3 Using inequality symbols

Time allocation 2 lessons

Suggested teaching and learning resources

Learners' experiences, checklist, braille.

Instructions

1 Let learners be in groups.
2 Ask learners to use inequality symbols in the following problems:
   a. The difference between 20 and 10 is less than the difference between 100 and 50
   b. when 1 is added to a certain number y the result is greater than 4
3 Let learners present their work.
4 Discuss with learners the answers as follows:
   a. $20 - 10 < 100 - 50$
   b. $1 + y > 4$
5 Let learners do exercise 28C from the learners' book.

Activity 4 Inserting inequality symbols in number sentences

Time allocation 1 lesson

Suggested teaching and learning resources

Number line, problem card, checklist.

Instructions

1 Let learners be in groups.
2 Distribute a card containing the following problem to each group.
   If $x > 4$ write the correct inequality symbol in each of the following:
   a. $3 \;\square\; x$
   b. $x + 2 \;\square\; 5$
3 Let learners present their work.
4 Discuss with learners to establish that,
   if $x > 4$ then
   a. $3 \;\boxed{<}\; x$
   b. $x + 2 \;\boxed{>}\; 5$
5 Let learners do exercise 28D from the learners book.

Review exercise

Time allocation 1 lesson

Let learners do the review exercise at the end of unit 28 in the learners' book.

Assessment

Use the table below to assess the learner.

Name of the learner: ______________________________

How best is the learner able to:	Very good	Good	Average	Needs support
1 complete number sentences?				
2 formulate number sentences?				
3 use inequality symbols appropriately?				

UNIT 29 Graphs

Time allocation 13 lessons

Introduction

Learners have been introduced to picture and bar graphs. They are able to draw and interprete graphs on a scale of 1:20. This unit will enable learners extend their knowledge as they will draw graphs up to a scale of 1:100. They will also be introduced to line graphs.

Success criteria

Learners must be able to:

- draw line graphs to scale up to 1:100
- interpret picture graphs
- interprete bar graphs
- interpret line graphs

Developmental areas

Skills

Ensure that learners develop the following skills:

- drawing line graphs
- interpreting line, bar and picture graphs

Concepts and knowledge

Ensure that learners acquire the concept and knowledge of graphs.

Attitudes and values

Ensure that learners appreciate the importance of graphs.

Special needs education

Ensure that the activities are adapted for learners with special educational needs. This will help them develop their potentials and become independent citizens. Observe and record their performance.

Background information

In this unit, learners will draw and interprete picture, bar and line graphs using a scale up to 1:100. It is also important that learners label both vertical and horizontal axes. When the scale is not given, learners should choose a scale which can easily be used for all numbers in a given data.

Activity 1 Collecting data

Time allocation 2 lessons

Suggested teaching, learning and assessment resources

Environment, charts, pental pens, checklist, braille.

Instructions

1 Ask learners to state their ages and record them.

2 Discuss with learners how to organise this data in a table eg:

Age (yrs)	10	11	12	13	14	15
No. of learners	4	6	13	5	3	2

3 Discuss with learners how tallies are used when collecting data.

4 Assist learners to appreciate the standard way of developing tallies.

5 Let learners be in groups.

6 Ask each group to collect data on one of the following: trees around the school, learners books on various learning areas, number of learners for surrounding villages and any other information which can be easily collected.

7 Ask each group to tabulate their data.

8 Let learners report their findings.

9 Let learners do exercise 29A from the learners' book.

Activity 2 Drawing picture graphs

Time allocation 2 lessons

Suggested teaching, learning and assessment resources

Data resources, progress report books, checklist, braille.

Instructions

1 Let learners be in groups.

2 Provide each group with the following data:

Number of trees planted by Zabwino school per year					
Year	2003	2004	2005	2006	2007
No. of trees	350	400	250	450	550

3 Let each group draw a picture graph to represent the information in the above table.

4 Let learners present their work.

5 Discuss with learners the graphs drawn.
6 Discuss the example in the learners book.
7 Let learners do Exercise 29B from the learners' book.

Activity 3 Drawing bar graphs

Time allocation 2 lessons

Suggested teaching, learning and assessment resources

Data resources, progress report books, graph paper, checklist, braille.

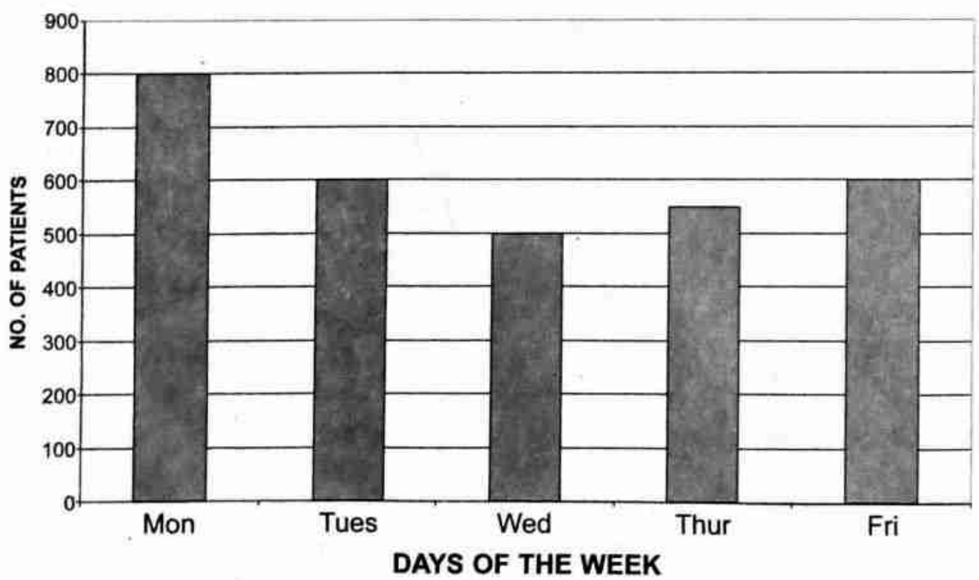
Instructions

1 Let learners be in groups.
2 Ask learners to draw a bar graph using the following information:

Days	Monday	Tuesday	Wednesday	Thursday	Friday
No. of patients	800	600	500	550	600

3 Let learners present their work.
4 Discuss with learners how to draw the bar graph as follows:

NUMBER OF PATIENTS IN 5 DAYS



5 Discuss the example in the learners' book.
6 Let the learners do Exercise 29C from the learners' book.

Activity 4 Drawing line graphs

Time allocation 2 lessons

Suggested teaching, learning and assessment resources

Data resources, grid board, ruler, checklist, braille.

Instructions

1 Let learners be in groups.
2 Ask them to discuss situations where line graphs are commonly used.
3 Discuss with learners how to draw a line graph using the following information.

Draw a line graph to represent mass of a baby in 6 months:

Time in months	1	2	3	4	5	6
Mass in kg	3	4	4.5	5	5.5	6.5

4 Let learners present their work.
5 Discuss with learners their findings.
6 Let the learners, in groups, discuss the example in their book.
7 Let the learners do Exercise 29D from the learners' book.

Activity 5 Interpreting picture graphs

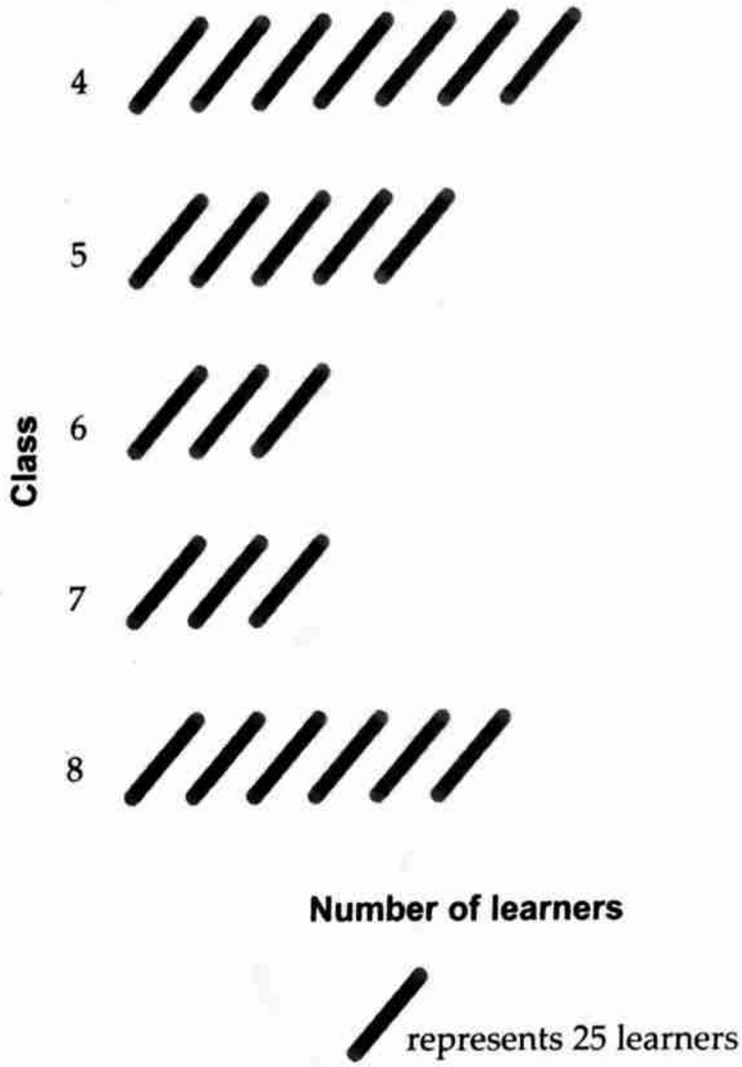
Time allocation 3 lessons

Suggested teaching, learning and assessment resources

Picture graphs, raised graphs, checklist, braille.

Instructions

1 Put up the following chart of a picture graph which shows the number of learners in Standards 4, 5, 6, 7 and 8 at Kale primary school.



2 Ask learners to identify the scale used in this graph.

3 Discuss with learners how to find the number of learners in each class as follows:

Number of learners in Standard 4	=	7 pictures x 25
	=	175
Number of learners in Standard 5	=	5 pictures x 25
	=	125
Number of learners in Standard 6	=	3 pictures x 25
	=	75
Number of learners in Standard 7	=	4 pictures x 25
	=	100

Number of learners in Standard 8 = 6 pictures x 25

= 150

4 Let learners, in groups, discuss the example in their books.

5 Let the learners do Exercise 29E from the learners' book.

Activity 6 Interpreting bar graphs

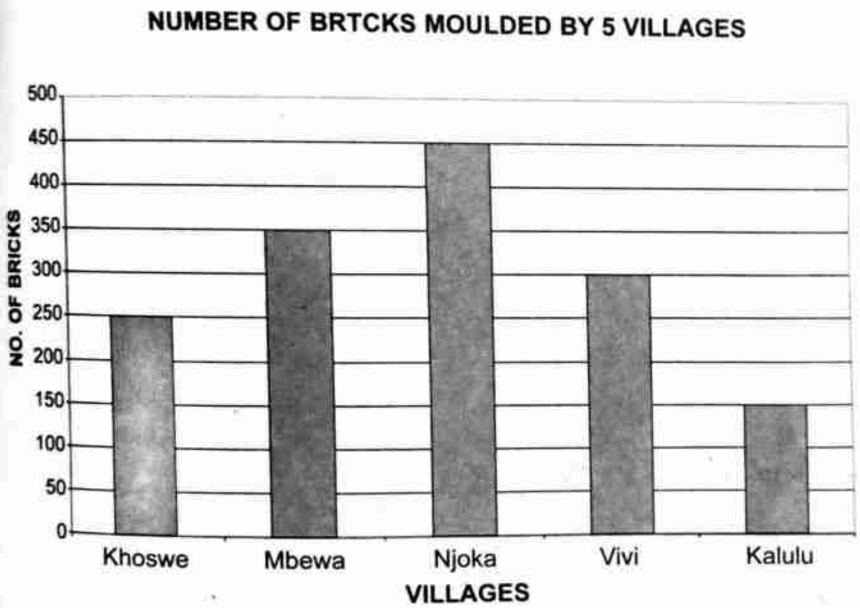
Time allocation 2 lessons

Suggested teaching, learning and assessment resources

Gridboard, bar graphs, raised graphs, checklist, braille.

Instructions

1 Display a bar graph which shows number of bricks moulded by 5 villages in one hour for construction of a classroom at a certain school.



2 Let learners identify the scale used in this graph.

3 Ask learners to discuss the number of bricks made by each village in one hour and the total of bricks made by all the 5 villages in one hour.

4 Discuss the example in the learners' book.

5 Let learners do Exercise 29F from the learners' book.

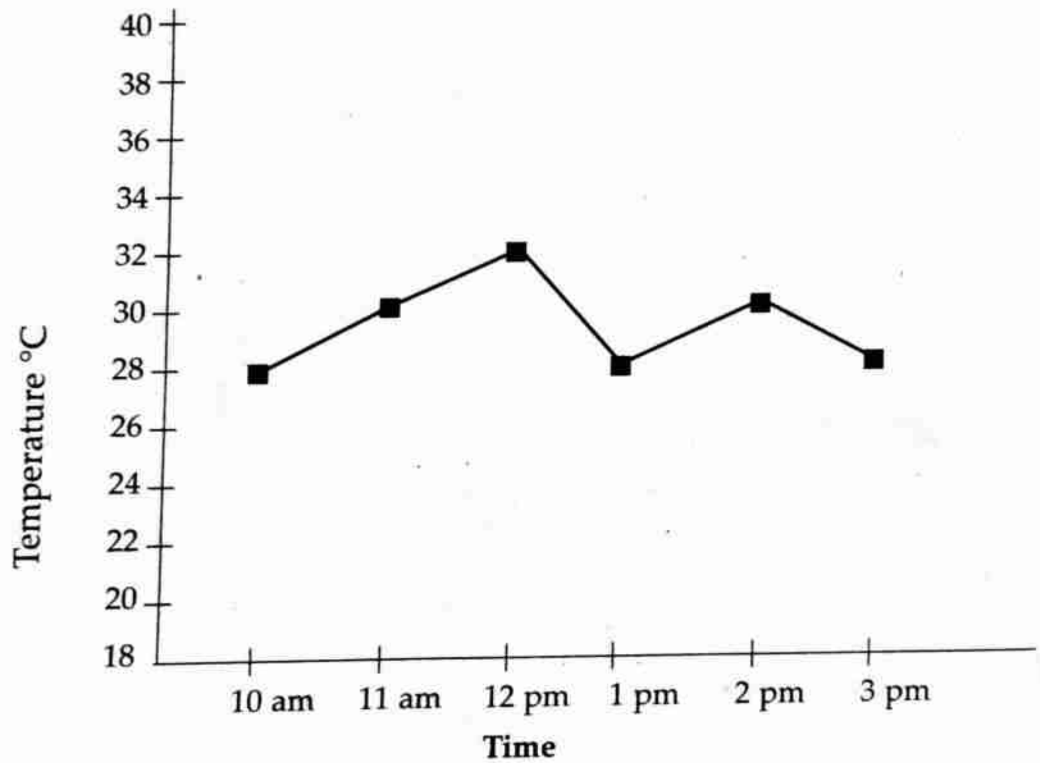
Activity 7 Interpreting line graphs

Time allocation 2 lessons

Suggested teaching, learning and assessment resource
Gridboard, line graphs, raised graphs, checklist, braille.

Instructions

1 Show learners the following line graph drawn on gridchart which represent temperature recorded at a certain weather station for six hours:



2 Ask learners to identify the scale used in the graph.

3 Discuss with learners how to use the line graph when answering the following questions:

a when was the heighest temperature recorded?

b what was the temperature at 11 am?

c at what times was the temperature the same

4 Let the learners do Exercise 29G from the learners' book.

Review exercise

Time allocation 1 lesson

Let learners turn to the learners' book and do the review exercise.

Assessment

Use the table below to assess each learner.

Name of the learner: ______________________________

How best is the learner able to:	Very good	Good	Average	Needs support
1 interpret picture graphs?				
2 interpret bar graphs?				
3 draw line graphs to scale up to 1:100?				
4 interpret line graphs?				

Answers

UNIT 1

Exercise 1A

1	60 000 000	70 000 000
2	85 000 000	90 000 000
3	22 600 000	52 600 000
4	67 000 000	74 000 000
5	48 000 000	46 000 000
6	25 000 000	22 500 000
7	80 600 000	76 600 000
8	95 400 000	94 300 000

Exercise 1B

1	500 000 000	600 000 000
2	650 000 000	800 000 000
3	210 000 000	215 000 000
4	800 000 000	700 000 000
5	120 000 000	180 000 000
6	498 000 000	497 000 000
7	250 000 000	325 000 000
8	665 000 000	660 000 000

Exercise 1C

1	40 000 000	50 000 000	60 000 000	70 000 000	80 000 000
2	285 000 000	291 000 000	295 000 000	327 000 000	365 000 000
3	650 000 000	700 000 000	750 000 000	800 000 000	850 000 000
4	101 500 000	102 000 000	102 500 000	103 000 000	103 500 000
5	40 000 000	42 000 000	45 000 000	47 500 000	50 900 000
6	801 407 000	801 700 000	802 100 000	802 205 000	802 400 000

Exercise 1D

1	295 000 000	270 000 000	190 000 000	175 000 000	85 000 000
2	1 000 000 000	904 000 000	892 000 000	708 000 000	586 000 000
3	835 000 000	830 860 210	740 990 125	740 990 050	725 642 001
4	816 002 000	670 100 000	505 340 000	403 000 019	185 750 025

5 88 500 000 | 88 050 000 | 87 500 000 | 87 060 000 | 87 005 000

6 1 000 000 000 | 999 999 997 | 989 000 800 | 930 450 000 | 903 054 600

5	88 500 000	88 050 000	87 500 000	87 060 000	87 005 000
6	1 000 000 000	999 999 997	989 000 800	930 450 000	903 054 600

Exercise 1E

1 Seventy eight million
2 One hundred and three million, eight thousand
3 Six hundred and seventy one million, five hundred and seven thousand, and eighty seven
4 four hundred and eighty three million, nine hundred and twenty one thousand, six hundred and sixty seven.
5 Eight hundred and seven million, two thousand and three
6 Nine hundred and eighty seven million, one hundred and eighty six thousand and one
7 Five hundred million, six hundred and two thousand and four.
8 Nine hundred and ninety nine million, nine hundred and ninety nine thousand, nine hundred and ninety seven.

Exercise 1F

1 709 400 000
2 680 020 007
3 902 722 000
4 369 618 549
5 841 000 038
6 1 000 000 000

UNIT 2

Exercise 2A

1 I, V, X
2 I, V, X
3 4, 7, 12

Exercise 2B

1 XV
2 XVI
3 XVIII
4 XIX
5 XX

Exercise 2C

1 6
2 11
3 20
4 18
5 19
6 4

Exercise 2D

1 I, IV, V, X, XII
2 II, IV, XI, XIX
3 III, V, VI, XIV, XIX
4 V, VI, VII, XI, XVIII, XX
5 I, VII, XIII, XVII

Exercise 2E

1 XVII, XIV, X, VI
2 XII, X, V, I
3 XVI, XIII, XI, VIII
4 XIX, XV, XII, IX, III
5 XX, XIX, XVIII, XI, IV

UNIT 3 Subtraction

Exercise 3A

1 1,804
2 2,629
3 19,487
4 1,500,000
5 52,725
6 15,402
7 16,816
8 3,634,838
9 4,135,949
10 251,270,100

Exercise 3B

1 4.48
2 0.58
3 190
4 201
5 2135
6 3105
7 536
8 625
9 2915
10 414

Exercise 3C

1 35,063
2 48,283
3 6,830,855
4 47,619
5 25,114,400
6 299,083
7 77,484
8 934,186
9 14,635,910
10 50,399

Exercise 3D

1 3440
2 2435
3 14
4 835
5 1729
6 22
7 14505
8 53
9 44
10 1897

Exercise 3E

1 52,846
2 7,528
3 318,825
4 16,260
5 316,460
6 51,709,584
7 16,151
8 683,381
9 6,189,212
10 3,610,818

Exercise 3F

1 11
2 201
3 31
4 73
5 72
6 173
7 200
8 14,008
9 436
10 718

Exercise 3G

1 1,290
2 382
3 641
4 2,138
5 15,705
6 10,126
7 4,281
8 2,128

Exercise 3H

1 ~~10,998~~ 9002
2 146
3 80
4 26
5 11,798
6 82,200 bags
7 1,150 pens
8 7,617 bags
9 510 learners
10 5 oranges

Review exercise

1 19,487
2 996
3 536
4 87 bags
5 105
6 784 bottles
7 500
8 146
9 6,215
10 501 learners

UNIT 4 averages

Exercise 4A

1 13
2 15
3 28
4 30
5 17
6 13
7 45
8 51.33
9 11.33
10 26

Exercise 4B

1 70
2 10
3 52
4 18
5 K63,000.00
6 32°C
7 62kg

UNIT 5 HCF and LCM

Exercise 5A

1 2
2 4
3 40
4 80
5 4
6 6

Exercise 5B

1 4
2 30
3 80
4 4
5 80

Exercise 5C

1 120
2 480
3 500
4 800
5 540
6 20

Exercise 5D

1 144
2 504
3 600
4 96
5 1,400
6 360
7 2,016
8 360

Exercise 5E

1 4
2 3
3 6
4 2,160
5 1,080
6 60

UNIT 6 Basic operations

Exercise 6A

1 $^{7}/_{24}$

2 $^{4}/_{21}$

3 $^{7}/_{15}$

4 $^{7}/_{48}$

5 $^{47}/_{60}$

6 $^{1}/_{3}$

7 $^{1}/_{48}$

8 $^{15}/_{16}$

Exercise 6B

1 $14^{11}/_{20}$

2 $6^{1}/_{18}$

3 $10^{15}/_{42}$

4 $2^{19}/_{30}$

5 $11^{1}/_{32}$

6 $9^{7}/_{8}$

7 $6^{3}/_{4}$

8 $7^{1}/_{2}$

9 $12^{3}/_{3}$

10 $15^{1}/_{12}$

Exercise 6C

1 $^{15}/_{16}$

2 $^{9}/_{20}$

3 $3^{1}/_{5}$

4 $2^{7}/_{34}$

5 $^{5}/_{6}$

6 $1^{3}/_{25}$

7 $^{4}/_{45}$

8 $^{16}/_{63}$

9 40

Exercise 6D

1 $^{17}/_{28}$

2 $^{3}/_{14}$

3 $^{17}/_{30}$

4 $^{19}/_{24}$

5 $7^{19}/_{30}$

6 $11^{1}/_{3}$

7 $49^{2}/_{3}$

8 $83^{5}/_{11}$

Exercise 6E

1 $^{32}/_{35}$

2 $^{4}/_{5}$

3 $1^{85}/_{88}$

4 $2^{3}/_{7}$

5 $2^{74}/_{204}$

6 $12^{7}/_{10}$

7 $2^{2}/_{3}$

8 $^{57}/_{64}$

9 $5^{2}/_{9}$

10 $1^{2}/_{3}$

11 $4^{22}/45$

Exercise 6F

1 $^{12}/_{35}$

2 $^{19}/_{21}$

3 $^{3}/_{20}$

4 $1^{1}/_{2}$

5 $^{4}/_{5}$

6 $2^{1}/_{2}$

Exercise 6G

1 $^{11}/_{28}$

2 $^{57}/_{78}$

3 $2^{3}/_{34}$

4 $^{5}/_{14}$

5 $^{4}/_{10}$

6 $5^{2}/_{15}$

7 $24^{13}/_{56}$

8 $3^{1}/_{4}$

9 12

10 21

Exercise 6H

1 $1^{3}/_{8}$

2 $5^{4}/_{5}$

3 $36^{1}/_{6}$

4 $3^{17}/_{26}$

5 $2^{4}/_{15}$

6 $1^{17}/_{20}$

7 $1^{1}/_{2}$

8 13 picture frames

UNIT 7 Basic operations

Exercise 7A

1 2.897

2 0.881

3 33.631

4 12.421

5 0.045

6 2.708

7 0.854

8 427.55

Exercise 7B

1 169.12
2 401.92
3 2,544.84 (2 dec. places)
4 974.211
5 0.408
6 126.1
7 180
8 11,520.576
9 19,421
10 210.3

Exercise 7C

1 1.109
2 49.4
3 30.31
4 1,058.86
5 728.91
6 108.718
7 2.25
8 62.07

Exercise 7D

1 1.004
2 2.3
3 607.24
4 3.89
5 1.17
6 26.76
7 3

Exercise 7E

1 0.751
2 10.4
3 20.19
4 1.89
5 33.88
6 0.007
7 6.4

Exercise 7F

1 0.91
2 93
3 8.593
4 0.205
5 62.505
6 0.2087
7 0.077

Exercise 7G

1 777.96
2 123.92 (2 dec. places)
3 649.6
4 3.2
5 15.048
6 3,024.55
7 11.926
8 5.25 bags

Review exercise

1 0.9625
2 23.811
3 57.294
4 399.18
5 11 dresses
6 3.5
7 5.09
8 58.1

UNIT 8

Exercise 8A

1 a. 109.55kg
  b. 0.06

2 **a.** 145.525
b. 0.126
c. 23.410
3 **a.** 26.8095
b. 36.0081
c. 90.3462m

Exercise 8B

1 $80^{1}/_{20}$
2 $750^{3}/_{40}$
3 $16^{6}/_{25}$
4 $30^{1}/_{500}$
5 $62^{63}/_{500}$
6 $2^{673}/_{1000}$
7 $3^{19}/_{50}$
8 $9^{247}/_{500}$
9 $1^{247}/_{2000}$
10 $15^{11}/_{250}$

Exercise 8C

1 12.88
2 10.67
3 2.17
4 2.22
5 20.33
6 8.67
7 15.43
8 11.57
9 48.42
10 42.83

Exercise 8D

1 7,000
2 4,600
3 7,140
4 10
5 1,160
6 94
7 50
8 0.457
9 0.026
10 0.00887

UNIT 9 Rates, ratio

Exercise 9A

1 49 km/h
2 31 km/h
3 a. 75 km/h, 120 km/h
b. 2.00 pm, 11.00 am
4 2.4 km/h

Exercise 9B

1 7:16
2 31:40
3 11:21
4 1:134
5 a. 13:12
b. 71:65
c: 71:60
d: 65:71:60
6. K75,000.00

Exercise 9C

1 James 32 biscuits
Jane 56 biscuits

Judith 24 biscuits

2 House 110,000 bricks

Kitchen 33,000 bricks

Toilet 22,000 bricks

3 Singano 624 learners

Mandala 520 learners

Nsigala 468 learners

4 a. 30 sweets

b. 72 sweets

Exercise 9D

1	Maize	21 bags
	Beans	42 bags
	Rice	63 bags
2	Luso	18kg
	Lennie	12kg
3	Chimwemwe	22 sweets
	Chifundo	33 sweets
	Mphatso	55 sweets
4	Bed	K2,000
	Table	K1,000
5	Pencil	K80
	Book	K640

Review exercise

1 15cm by 20cm by 25cm

2 28 days

3 420 men

4 Thema 24, Godfrey 36, Tina 45 sweets

5 Thoko 120, Joyce 90, Stanley 160 oranges

6 3.3 km/minute

7 85.7 km/hr

UNIT 10 Percentages

Exercise 10A

1 $^{4}/_{5}$

2 $^{47}/_{100}$

3 $^{24}/_{25}$

4 $^{1}/_{50}$

5 $^{9}/_{50}$

6 $1^{1}/_{5}$

7 $^{3}/_{50}$

8 $1^{47}/_{100}$

9 $^{21}/_{25}$

10 $^{1}/_{2}$

Exercise 10B

1 20%

2 75%

3 4%

4 150%

5 133.33%

6 866.67%

7 80%

8 2025%

9 15%

10 17%

Exercise 10C

1 0.41

2 0.28

3 1.5

4 0.99

5 0.8

6 2.65

7 2.75

8 0.11

9 0.07

10 2.11

Exercise 10D

1 33%

2 64%

3 283%

4 930%

5 620%

6 158%

7 95%

8 1%

9 341%

10 40%

Exercise 10E

1 75%

2 400%

3 18 marks

4 17,129

5 K10,350,000.00

Review exercise

1 a. $^{29}/_{50}$

b. $2^1/_4$

2 a. 250%

b. 40.06%

3 a. 5.13

b. 0.79

4 a. 46%

b. 112%

5 175%

UNIT 11 Bills and budgets

Exercise 11A

1 K388.00

2 K1,656.00

3 K891.25

4 K683.30

5 K107.50

Exercise 11B

1 K1,823.00

2 K3,282.90

3 K398.95

4 K1,074.30

5 K2,178.00

Review exercise

1 K640.00

2 K1,058.00

3 K399.50

4 K552.50

5 K660.00

UNIT 12 Profit and loss

Exercise 12A

1 CP= K3,560.00

SP = K3,500.00

2 SP = K966,500.00

CP= K980,000.00

3 CP= K36,000.00

SP = K45,000.00

4 CP= K225.00

SP = K250.00

Exercise 12B

1 a. Loss = K83.10

b. Profit = K55,148.64
c. Loss = K81,000.00
d. Profit = K1,500,000.00
e Loss = K35,250.00

2 K1,400.00
3 K369,000.00
4 K6,735.00
5 K375,000.00

Exercise 12C

1 a. K2,713.00
b. K993,005.75
c. K2,261,509.85
d. K10,492,020.00

2 K1,707.00
3 K109,250.00

Exercise 12D

1 a. K12,751.50
b. K72,122.70
c. K84,788.96
d. K964,271.00

2 K7,220.00
3 a. K5,350.00 ✓
b. K642,000.00 ✓

Review exercise

1 a. Profit = K36.00
b. SP = K23,295.00
c. CP= K79,311.22
d. SP = 3600t
e. CP= K1,853,850.00
f. SP = K10,745,000.00

2 a. K9,763.00
b. K10,199.00
c. K147,753.00
d. She made a loss

UNIT 13 Commission

Exercise 13A

1 a. K28.00
b. K81.00
c. K5.00
d. K152.00
e. K768.00

2 K1,500.00
3 K50.00
4 K3,500.00
5 K150.00

Exercise 13B

1 K1,472.00
2 K1,886.00
3 K973.00
4 K13,060.00
5 K6,300.00
6 K796.20

Exercise 13C

1 K280.00
2 K52.50
3 K1,000.00
4 K69,200.00
5 K800.00
6 K34.50
7 K1,600.00

UNIT 14 Postal services

Exercise 14A

1 a. K40.00
b. K40.00
c. K195.00
d. K370.00

2 a. K100.00
b.(i) K25.00
(ii) K240.00
3 a. K0.00
b. K0.00
4 K195.00
5 K395.00

Exercise 14B

1 K1,055.00
2 K2,775.00
3 K1,055.00
4 a. K220.00
b. K155.00
5 K2,645.00

Exercise 14C

1 a. K5.00
b. K300.00
c. K250.00
2 a. K660.00
b. K1,650.00
3 K100.00
4 K4,400.00
5 K150.00
6 K670.00
7 K3,365.00
8 K835.00
9 K2,265.00
10 K2,265.00

Exercise 14D

1 a. K55.00
b. K150.00
c. K450.00
2 K410.00
3 K300.00
4 K2,070.00
5 K4,390.00
6 a. K675.00
b. K1,435.00
c. K5,017.00
7 K4,975.00
8 K4,435.00
9 K1,615.00
10 K765.00

Exercise 14E

1 a. K920.00
b. K1,670.00
2 K3,500.00
3 K3,850.00
4 K1,171.00
5 K2,410.00

Exercise 14F

1 K96.00
2 K111.00
3 K126.00
4 a. K63.00
b. K75.00
5 a. Practical
b. K60.00

UNIT 15 Bank services

Exercise 15A

1 National Bank, Standard Bank, Inde Bank, New Building society Bank (NBS Bank) Malawi Savings Bank Opportunity Bank and First Merchant Bank.
2 i Keeping Money safe
ii Lending Money

iii Facilitating payment of bills

3 Practical

4 Practical

5 Practical

6 Practical

7 Practical

8 Practical

UNIT 16 Simple interest

Exercise 16A

1 K600

2 K24

3 K6,300

4 K1,080

5 K2,240

6 K1,600

Exercise 16B

1 a. K3,600

b. K11,500

c. K52,000

2 K28,750

3 K55,000

Exercise 16C

1 a. 10%

b. $^{5}/_{6}$%

c. 10%

d. 1%

2 $6^{2}/_{3}$%

3 2%

Exercise 16D

1 10 months or $^{5}/_{6}$ yrs

2 $^{1}/_{2}$ yr or 6 months

3 $1^{1}/_{4}$ yrs

4 2 yrs

Exercise 16E

1 K10,000

2 K2,000

3 K100,000

4 K7,291.67

5 K40,000

Review exercise

1 K80

2 K1,800

3 K11,475

4 K31,312.50

5 K1,512

6 K360,000

7 K315,000

8 K8,000

9 1 year

10 $^{1}/_{8}$ yr

11 2.8%

UNIT 17 Cash and bank account

Exercise 17A Cash account

1 MPONDA ~~Kaphut~~a school cash account

Dr | Cr

Date	Income	Amount	Date	Payments	Amount
		K t			K t
1-01-08	Cash in hand	6,500 00	2-01-08	Bought pens	2,000 00
2-01-08	Fundraised	2,750 00	9-01-08	Bought books	3,125 00
27-01-08	Sold trees	1,890 00	26-01-08	Paid wages	2,200 00
			31-01-08	Balance c/d	3,815 00
		11,140 00			11,140 00
31-01-08	Balance b/d	3,815 00			

2 Chikondi's cash account

Dr | Cr

Date	Income	Amount	Date	Payments	Amount
		K t			K t
1-07-08	Capital	600 00	3-07-08	Bought sugarcane	570 00
5-07-08	Sold sugarcane	450 00	13-07-08	Bought sugarcane	600 00
9-07-08	Sold sugarcane	150 00	31-07-08	Balance c/d	930 00
12-07-08	Sold sugarcane	500 00			
17-07-08	Sold sugarcane	400 00			
		2,100 00			2,100 00
31-07-08	Balance b/d	930 00			

3 Wild Life club cash account

Dr Cr

Date	Income	Amount	Date	Payments	Amount
		K t			K t
1-03-08	Cash in hand	7,000 00	3-03-08	Bought chicks	2,400 00
29-03-08	Sold chickens	5,500 00	3-03-08	Bought feed	850 00
			3-03-08	Bought parrafin	400 00
			14-03-08	Paid water bill	650 00
			23-03-08	Bought feed	2,200 00
			27-03-08	Paid water bill	550 00
			31-03-08	Balance c/d	5,450 00
		12,500 00			12,500 00
31-03-08	Balance b/d	5,450 00			

4 Temwa's cash account

Dr Cr

Date	Income	Amount	Date	Payments	Amount
		K t			K t
1-12-08	Capital	3,000 00	2-12-08	Bought watches	1,800 00
6-12-08	Received cash	1,700 00	9-12-08	Bought spare parts	950 00
8-12-08	Sold watch	1,750 00	25-12-08	Bought spart parts	1,800 00
14-12-08	Received cash	1,120 00	31-12-08	Balance c/d	4,970 00
20-12-08	Sold watch	1,950 00			
		9,520 00			9,520 00
31-12-08	Balance b/d	4,970 00			

Exercise 17B

1 Agriculture club's bank account

Dr Cr

Date	Income	Amount	Date	Payments	Amount
		K t			K t
1-01-08	Bank cash	10,500 00	3-01-08	Bought seeds	950 00
11-01-08	Bank cheque	2,500 00	3-01-08	Bought watercan	750 00
25-01-08	Deposited cheque	3,000 00	20-01-08	Bought fertilizer	3,700 00
			21-01-08	Withdrawal cash	1,500 00
			31-01-08	Balance c/d	8,600 00
		15,500 00			15,500 00
31-01-08	Balance b/d	8,600 00			

2 Chifundo's bank account

Dr Cr

Date	Income	Amount	Date	Payments	Amount
		K t			K t
1-03-08	balance in hand	8,225 00	2-03-08	bought materials	2,000 00
4-03-08	banked cash	3,000 00	9-03-08	bought materials	3,125 00
14-03-08	banked cheque	5,760 00	26-03-08	transport	2,200 00
27-03-08	deposited cheque	2,700 00	20-03-08	paid wages	2,440 00
			27-03-08	transport	1,900 00
			31-03-08	Balance c/d	9,945 00
		19,685 00			19,685 00
31-03-08	Balance b/d	9,945 00			

3 Mr Mbatata's bank account

Dr					Cr
Date	Income	Amount	Date	Payments	Amount
		K t			K t
1-09-08	balance in hand	12,000 00	2-09-08	purchased goods	5,100 00
15-09-08	banked cash	3,950 00	4-09-08	paid for transport	1,800 00
26-09-08	banked cheque	4,550 00	20-09-08	purchased goods	4,530 00
			24-09-08	stationery	950 00
			27-09-08	paid wages	3,280 00
			30-09-08	Balance c/d	4,840 00
		20,500 00			20,500 00
30-09-08	Balance b/d	4,840 00			

4 Mrs Sikono's bank account

Dr					Cr
Date	Income	Amount	Date	Payments	Amount
		K t			K t
1-10-08	balance in bank	9,740 00	3-10-08	for stationery	1,200 00
7-10-08	banked cash	5,610 00	5-10-08	bought goods	3,500 00
20-10-08	banked cash	2,350 00	15-10-08	bought goods	4,800 00
22-10-08	banked cheque	6,170 00	27-10-08	paid wages	2,400 00
			31-10-08	Balance c/d	11,970 00
		23,870 00			23,870 00
31-10-08	Balance b/d	11,970 00			

5 Chimwemwe's bank account

Dr | Cr

Date	Income	Amount	Date	Payments	Amount
		K t			K t
1-04-08	balance in bank	10,500 00	2-04-08	bought clothes	5,100 00
5-04-08	deposited cash	3,870 00	7-04-08	paid for transport	1,500 00
12-04-08	deposited cash	4,500 00	20-04-08	bought shoes	4,530 00
22-04-08	deposited cash	4,900 00	27-04-08	paid rent	1,750 00
24-04-08	deposited cheque	2,400 00	27-04-08	paid wages	2,500 00
			30-04-08	Balance c/d	9,970 00
		26,170 00			26,170 00
30-04-08	Balance b/d	9,970 00			

Review exercise

1 Mathematics club cash account

Dr | Cr

Date	Income	Amount	Date	Payments	Amount
		K t			K t
1-01-08	contributions	2,350 00	2-01-08	bought stationery	950 00
5-01-08	fundraised	3,000 00	8-01-08	paid transport	2,150 00
20-01-08	received donation	5,000 00	8-01-08	paid for drinks	600 00
			15-01-08	paid instruments	1,745 00
			27-01-08	bought books	2,500
			31-01-08	Balance c/d	2,405 00
		10,350 00			10,350 00
31-01-08	Balance b/d	2,405 00			

2 Maria's cash account

Dr Cr

Date	Income	Amount	Date	Payments	Amount
		K t			K t
1-08-08	Capital	4,000 00	2-08-08	bought spare parts	2,000 00
5-08-08	received cash	1,750 00	14-08-08	bought spare parts	3,125 00
9-08-08	received cash	3,300 00	27-08-08	paid rent	1,500 00
17-08-08	received cash	2,000 00	28-08-08	paid transport	750 00
20-08-08	sold a watch	1,900 00	31-08-08	Balance c/d	6,445 00
		11,140 00			12,950 00
31-08-08	Balance b/d	3,815 00			

3 A school's bank account

Dr Cr

Date	Income	Amount	Date	Payments	Amount
		K t			K t
1-08-07	balance in bank	6,500 00	2-08-07	bought books	3,420 00
4-08-07	banked	2,750 00	9-08-07	bought cement	5,000 00
12-08-07	banked cheque	1,890 00	26-08-07	paid builders	2,320 00
27-08-07	deposited cheque	1,900 00	31-08-07	Balance c/d	19,460 00
		30,200 00			30,200 00
31-08-07	Balance b/d	19,460 00			

4 Tawona dairy farmers association's bank account

Dr Cr

Date	Income	Amount		Date	Payments	Amount	
		K	t			K	t
1-07-08	balance in bank	20,000	00	2-07-08	bought feed	3,200	00
6-07-08	banked	3,500	00	9-07-08	paid for vaccines	4,050	00
12-07-08	deposited cash	6,380	00	14-07-08	bought milk cans	2,170	00
19-07-08	deposited cash	7,210	00	20-07-08	" animal feed	2,300	00
24-07-08	" cheque	1,700	00	25-07-08	paid water bill	2,210	00
28-07-08	" cheque	5,400	00	31-07-08	Balance c/d	30,260	00
		44,190	00			44,190	00
31-07-08	Balance b/d	30,260	00				

UNIT 18 Geometric shapes

Exercise 18

1

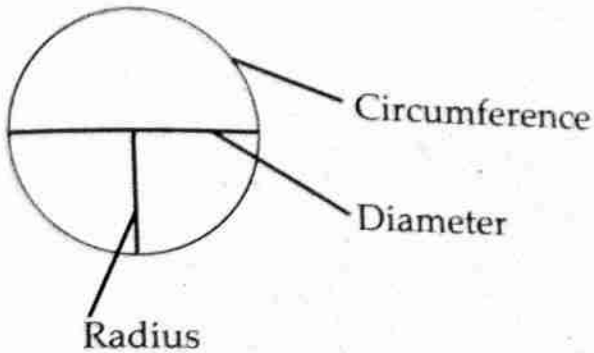
Type of triangle	Properties
Equilateral	All sides are equal All angles are equal
Scalene	All sides different All angles different

2 **a.** Rectangle
 b. Parallelogram
 c. Rhombus
 d. Square

3 The diffeerences are:

a. Parallelogram opposite sides are equal while rhombus has all sides equal

b. A square has all sides equal while a rectangle has opposite sides equal

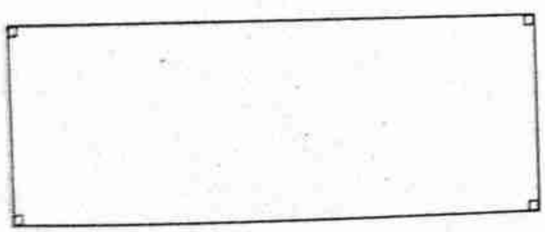


Review exercise

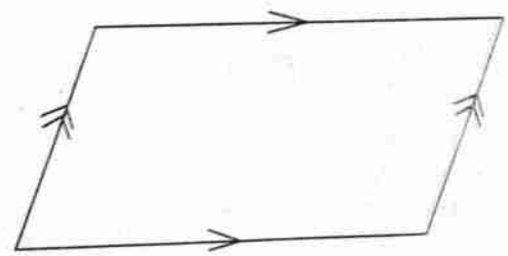
1 Equilateral isosceles and scalene triangles

2 a. Equilaterals triangle
  b. scalene triangle
  c. Isosceles triangle
  d. Scalene triangle

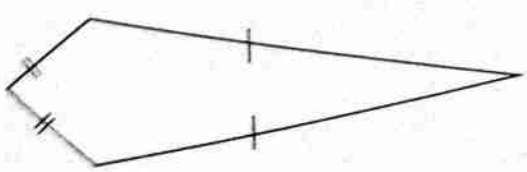
3



Rectangle



Parallelogram



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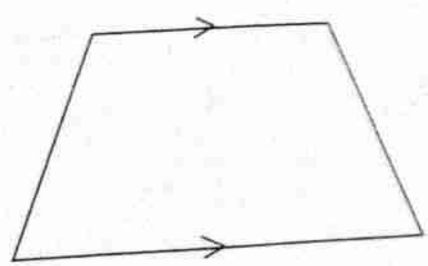
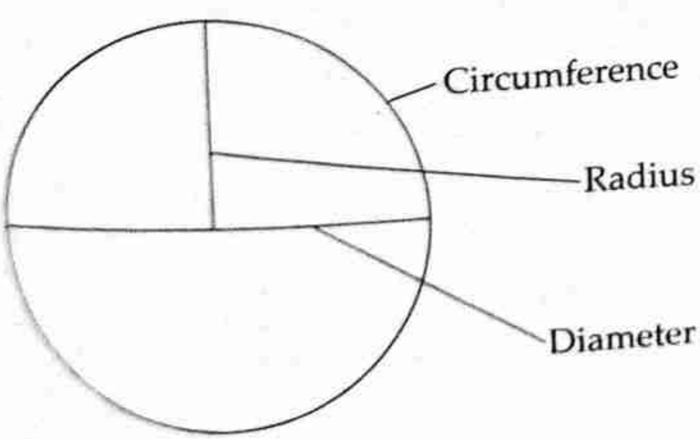


Trapezium



UNIT 19 Construction of angles

Exercise 19A

1 80°
2 50°
3 130°
4 300°
5 130°
6 300°

Exercise 19B

Practical

Exercise 19B

1 K952.90
2 K1 322.75
3 a K12 029.10 b K470.90

Review exercise

Practical

UNIT 20 Length

Exercise 20A

1a. 22cm b. 132cm
c. 25.142cm or $25^1/_7$cm

2a. 56.57cm or $56^4/_7$ b. 11m

3 66 metres
4 37.68cm
5 24.5cm
6 21cm

Exercise 20B

1 63cm
2 32cm
3 60cm
4 70cm
5 62cm
6 37cm

Review exercise

1 56.57cm
2 62.85cm
3 9.43cm
4 4.4cm

UNIT 21 Mass

Exercise 21A

1 785kg 253gm
2 222.187kg
3 348kg 625gm
4 1,587.938kg
5 5,879kg 763g
6 7,314.36kg
7 90kg
8 14.53kg

Exercise 21B

1 2,615kg
2 52 cartons
3 80kg
4 150kg
5 201,500kg
6 11 trips
7 1,028.55kg
8 4.534kg

Review exercise

1 408kg 770g
2 235kg 51g
3 14.11kg
4 37kg 300g
5 **a.** 168.5kg
b. 33.7kg

UNIT 22

Exercise 22A

1 $2^{3}/_{14}$ fortnights
2 16 years old
3 4 human generations
4 2,020
5 2 decades
6 Yankho
7 9 decades
8 $6^{1}/_{2}$ fortnights
9 35 years
10 4 silver-jibilees

Exercise 22B

1a. 70 years old **b.** 77 years old
2 practical
3 practical
4 30 centenaries
5 78 years
6 3,000

Exercise 22C

1 39 years
2 93 years
3 1,793 years
4 3,605 years
5 14 years

Review exercise

1 42 BC
2 The man
3 49 BC
4 1989 AD
5 2062 AD

1993
+ 200
2193
199
− 2
9

UNIT 23 Area

Exercise 23A

1 42cm^2
 b. 70cm^2
 c. 48cm^2
 d. 88cm^2
2 60cm^2
3 350m^2
4 11cm

Exercise 23B

1 a. $254^4/_7$cm^2
 b. $50^2/_7$cm^2
 c. $346^1/_2$cm^2
2 a. $12^4/_7$cm^2
 b. $9^5/_8$cm^2
3 154m^2
4 $201^1/_7$cm^2

Exercise 23C

1 317m^2
2 140cm^2
3 43m^2
4 514m^2
5 67m^2
6 $10^1/_2$m^2

Review exercise

1 15cm^2
2 4cm^2
3 8cm^2
4 a. 280cm^2
 b. 205cm^2
5 $1^{13}/_{14}$m^2

UNIT 24 Volume and capacity

Exercise 24A

1 32l
2 0.475l
3 25,850cm^3
4 628ml
5 340,000cm^3
6 0.875l
7 950cm^3
8 78.54ml
9 690cm^3
10 1,200

Exercise 24B

1 30cm^3
2 960cm^3
3 560cm^3
4 1250cm^3
5 240cm^3